A Review of Advancements and Trends in Time-to-Digital Converters Based on FPGA

Haojie Xia, *Member, IEEE*, Xin Yu, Jin Zhang, and Guiping Cao

***Abstract*— Recently, advancements have been made in the design, implementation, and application of time-to-digital converters (TDCs) based on field-programmable gate array (FPGA) technology. The progress can be attributed to the low cost, short development cycle, and easy integration offered by FPGA platforms, which continuously provide TDCs with state-of-the-art high-performance hardware featuring low latency, low jitter, and multiple interfaces. Moreover, the progress has been driven by the demand for high-performance TDCs tailored to specific applications. The demand has driven the improvement, upgrading, and iterative optimization of various modules within TDC systems. Therefore, we present a comprehensive review and summarize the research reports on FPGA-based TDCs. The aim of the review is to clarify the ongoing efforts to improve measurement resolution and precision, reduce nonlinearity, and enhance the reliability of TDCs in specific applications. This review summarizes and analyzes the development of FPGA-based TDCs from both technical and application perspectives to consolidate past research findings and outline the directions for future research. This article reviews the latest literature on the implementation architecture and performance of FPGA-based TDCs, refining the classification methods for implementation architectures and summarizing the performance evaluation metrics. Through comparisons and analysis, this review identifies the limitations faced by FPGA-based TDC research, highlights pressing issues that need to be addressed, and explores potential challenges that may arise.**

***Index Terms*— Field-programmable gate array (FPGA), measurement uncertainty, time-interval measurement, time-to-digital converter (TDC).**

I. Introduction

TIME-TO-DIGITAL converters (TDCs) are devices used to measure precise time intervals. Generally, TDCs are employed to measure time intervals in the nanosecond range or smaller. TDCs have been widely used in the field of physical research, particularly in applications based on the time-of-flight (ToF) method. TDCs are extensively used in nuclear physics in various studies involving TDC-based applications. For instance, in nuclear physics experiments, the coupling of photomultiplier tubes and fast scintillators generates numerous pulses, and the TDC plays a crucial role as an integral component of the pulse processing circuit. In medical imaging, TDCs are frequently utilized in positron emission tomography (PET) systems to measure the different flight times of human tissues or substances and generate medical images based on the measurement results. In scientific instruments, such as mass spectrometers, TDCs are often employed to measure the flight time of ions. When test atoms or molecules are ionized and stripped of a corresponding number of electrons to produce positive ions, they are accelerated and ejected over a fixed distance. Subsequently, ions with different masses have different velocities, resulting in varying flight times. Information pertaining to the mass-to-charge ratio of the test atoms or molecules can be obtained by measuring the changes in flight time using TDCs. TDCs are used extensively in light detection and ranging (LiDAR). In LiDAR, when a laser emits light to illuminate a target, it triggers the TDC to begin measurement. When the receiving device detects the light reflected from the target, the measurement stops and the result is output. This distance can be determined by multiplying the measured flight time of the photons with the speed of light. In summary, the principle of TDC in these applications primarily involves measuring the time interval between the emission and reception signals to reveal information on the object being tested or the environment in which the signals propagated via reflection.

The earliest TDCs were implemented using analog circuits. Generally, scholars have utilized a time-to-voltage converter (TVC) combined with an analog-to-digital converter (ADC) to convert a time measurement into an analog signal and then into a digital quantity [1], [2], [3]. However, such analog circuits have complex structures, large sizes, and high power consumption, which severely limit their applications. With the advancement of semiconductor technology, digital-circuit-based TDCs have emerged. Researchers have used digital counters or flash memory to fabricate all-digital TDCs based on application-specific integrated circuits (ASICs). Digital TDCs can achieve a considerably larger dynamic measurement range than analog TDCs. However, they are still constrained by the reference clock frequency, and the minimum achievable

Manuscript received 2 November 2023; revised 24 January 2024; accepted 26 January 2024. Date of publication 26 June 2024; date of current version 4 July 2024. This work was supported in part by the National Key Research and Development Program of China under Grant 2023YFF0719700 and in part by the National Natural Science Foundation of China under Grant 51975179. The Associate Editor coordinating the review process was Dr. Alvaro Hernandez. *(Corresponding author: Haojie Xia.)*

Haojie Xia, Xin Yu, and Jin Zhang are with Anhui Province Key Laboratory of Measuring Theory and Precision Instrument, School of Instrument Science and Optoelectronics Engineering, Hefei University of Technology, Hefei, Anhui 230009, China (e-mail: hjxia@hfut.edu.cn; yuxin@mail.hfut.edu.cn; zhangjin@hfut.edu.cn).

Guiping Cao is with China Research Center of High Speed Machine Vision, Hefei I-TEK OptoElectronics Company Ltd., Hefei, Anhui 230088, China (e-mail: caoguiping@i-tek.cn).

Digital Object Identifier 10.1109/TIM.2024.3419091

delay is determined by gate delays dictated by the manufacturing process. Researchers have made numerous attempts to achieve subgate delay resolution in TDCs but have achieved limited success. It was not until the emergence of digital circuit-based delay chains, phase-locked loops (PLLs), and delay-locked loops (DLLs) that researchers achieved time steps smaller than gate delays. With advancements in technology, TDCs are being increasingly applied and developed in various fields, and researchers have designed various architectures to implement TDCs and have realized targeted improvements in terms of measurement resolution, precision, and reliability. These include novel TDC architectures realized using techniques such as time stretching, Vernier differential, pulse shrinking, gated-ring oscillators, and successive approximations. These techniques can be combined to achieve a subpicosecond resolution.

In terms of hardware implementation, the modern TDC architecture is primarily based on two platforms: ASIC and field-programmable gate array (FPGA). Continuous development of FPGA technology has gradually narrowed the performance gap between FPGA and ASIC. This is primarily reflected in the improvement of FPGA's logic resource capacity, inherent internal delay, and power consumption, which enables TDC to achieve the same level of measurement resolution and accuracy as ASIC platforms on high-performance FPGAs with 28- or 20-nm processes [58], [82], [157]. In addition, developing applications based on FPGA platforms has obvious advantages, such as high flexibility, low cost, and short research and experimental development (R&D) cycles. Therefore, an increasing amount of research on TDC has focused on the FPGA platform. Significant differences exist in the research focus between TDCs developed based on FPGA and those developed based on ASIC. Most of TDCs based on FPGA are used as precision time measurement modules in specific applications, and thus, they focus on the resolution and sampling rate of TDC. The time-interval characteristics required for measurement differ based on the different application scenarios; therefore, modular designs must be personalized based on the system requirements. If researchers focus more on the TDC technology itself, they will focus on improving measurement resolution, linearity, and reliability. The improvement in these performance parameters will be based on a large amount of experimental statistical data. Owing to the customized architecture and building elements used in ASIC-based TDCs, regardless of where the TDC is applied, this research provides a detailed introduction to each component and analyzes their performance based on experimental data. In summary, the differences are primarily caused by the high dependence of the TDC on the hardware used to implement its architecture. By using ASIC, the TDC architecture can be customized, while on an FPGA with predefined logic elements and routing resources, it is necessary to study the most suitable specific applications for TDC implemented by specific hardware and architecture and discover the limitations of FPGA devices and TDC implementation architectures for specific applications to outline future development directions.

A TDC based on an FPGA can achieve a time measurement resolution of subpicoseconds. Wu and Shi [4] proposed the wave union (WU) method, which uses a single delayed chain structure to measure the test signal multiple times to subdivide the ultrawide bins, ultimately achieving a root-mean-square (rms) resolution of 10 ps. Subsequent research achieved measurement resolutions below 10 ps [5], [6], [7], [8], [9], [10], [11], [12], [13], with Wang et al. [14] achieving an ultrahigh resolution of 0.46 ps. With the continuous optimization of the latest FPGA device architecture, increasing resources, and improving the layout, it has become easier to implement the multichain tapped delay line (TDL) architecture and can be combined with more advanced sampling and calibration methods. Without considering the balance of resources and power consumption, the multichain architecture can achieve significantly improved resolution and lower nonlinearity compared to the single-chain architecture.

To improve the measurement resolution and linearity of TDCs in specific application scenarios, several recent literature reviews have summarized the latest progress in the architecture, implementation methods, and performance of TDCs based on ASIC platforms [15], [16], [17]. However, few comprehensive literature reviews have focused specifically on FPGA-based TDCs. Machado et al. [18] conducted a comprehensive classification and summary of these works. In this article, we present a comprehensive review of related research work, including our preliminary work, to cover the relevant research conducted in recent years on FPGA-based TDCs and better summarize the optimal performance achievable by FPGA-based TDCs under different architectures and implementation schemes and discuss the problems associated with these architectures and methods to find possible solutions. The ultimate goal is to identify solutions to existing problems, explore potential development directions for FPGA-based TDCs, and uncover potential applications that have not been considered in the current research.

This study contains several parts. In Section II, the architectures or methods adopted by various modules of the FPGA-based TDCs are classified and summarized, including the main achievements of correlational research. Section III analyzes the main performance indicators of an FPGA-based TDC and their influencing factors. Section IV discusses the main problems and challenges faced in the design, implementation, and evaluation of FPGA-based TDCs based on the insights gained from the literature review. In addition, potential solutions are proposed for the future and directions for future research are highlighted. Finally, this article is summarized in Section V.

II. Architectures and Implementation

With the increasing number of studies on FPGA-based TDCs, it is necessary to establish a unified classification standard to facilitate future research advancement and summarization. Previous studies have been primarily classified based on architecture; however, this approach has resulted in numerous similar architectures being assigned different names, causing confusion. We found that regardless of the architecture used for implementing a TDC system, the final TDC system inevitably comprises signal input, coarse counter, fine-time interpolation, decoder, and calibration modules, as shown in Fig. 1. When implemented on an FPGA, the hit signal enters

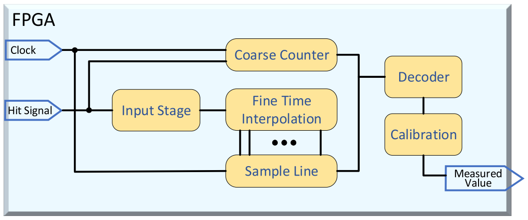

Fig. 1. Block diagram of an FPGA-based TDC.

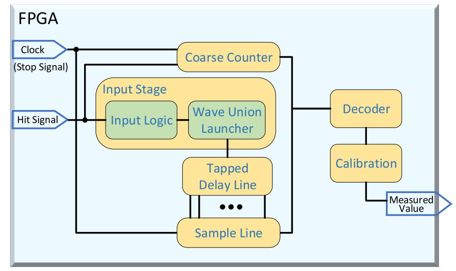

Fig. 2. Block diagram of an FPGA-based TDC using the WU method.

the TDC through the signal input stage. The coarse counter module counts the integral period of the time interval to be tested, while the fine-time interpolation module subdivides time intervals smaller than one reference clock period based on the designed architecture to complete fine counting. The outputs of the coarse counter and fine-time interpolation modules are then sampled and the decoder completes the binary conversion of the code. A dedicated nonlinear calibration module is designed to calibrate the measurement nonlinearity caused by factors such as architecture, layout and routing, manufacturing processes, power supply voltage, and temperature (PVT). All of these modules mentioned above are indispensable in the implementation of TDC, and different studies have reported different results because they differ in the implementation of certain modules compared to others. Therefore, we classified and summarized the architectures and methods employed on these modules in different studies.

A. Signal Input Module

The input module of the TDC refers to the module that handles the input signals, such as the measured signal, system clock, and reference clock, at the input end of the FPGA-based TDC. In practical applications, various auxiliary circuits are incorporated into the input module to suppress noise in the input signals and achieve pulse uniformity in the measured signal. In certain studies, authors have focused on reducing the resource consumption and addressing the dead-time problem of at least two clock cycles caused by the inability to input a second measured signal within the same clock cycle [19], [20]. As an integral component of the TDC, improvements in the signal input module can enhance the resolution and reduce nonlinearity. For instance, the WU method proposed by Wu and Shi [4] introduced a WU launcher into the signal input stage. It generated multiple edges for the measured signal and fed them into a fine-time interpolation module with a single delay chain. By performing multiple measurements using a single delay chain, it subdivides ultrawide bins (a series of buffer zones constituting the TDL).

As shown in Fig. 2, the WU method operates within a delay-chain architecture, enabling multiple measurements of a single delay chain by altering the number of signal edges at the input stage using a WU launcher. Ultimately, it resulted in in a resolution with an rms value of 10 ps. Based on the launcher used, the WU method can be further classified as the WU-A scheme, which utilizes a finite step response launcher with two available logic inversions, or the WU-B scheme, which utilizes an ISR launcher based on a ring oscillator. The difference between the two schemes lies in their finite and infinite pulse responses. The WU-A scheme requires the inclusion of a calibration module in the TDC to improve the measurement precision and resolution, whereas the WU-B scheme achieves a 10-ps rms measurement resolution without requiring calibration.

Since the introduction of the WU method, it has been widely used in TDCs utilizing delay chains as a fine interpolation method and has been improved for higher measurement resolution, lower nonlinearity, and better reliability [21], [22], [23], [24]. Wang et al. [5] proposed a multiple time-averaged TDC. Their work integrated an improved WU method into the delay chain, achieving fourfold time averaging and a 9-ps rms resolution. In addition, the number of channels was expanded to nine, demonstrating that the use of the WU method in the delay-chain architecture not only improves the resolution but also enhances the reliability of the multichannel TDC. Liu and Wang [9] implemented a WU-TDL architecture TDC with 128 channels using on an FPGA the latest advanced manufacturing process. They proposed that the measurement precision could be further improved to 3.1 ps by merging multiple TDC blocks [25]. This WU-TDL architecture-based TDC was implemented in Ultrascale and ZYNQ series FPGA devices and achieved a measurement resolution and precision better than 10 ps [12], [24]. Through continuous research and experimental comparisons on different FPGA devices, it has been found that the WU-TDL architecture not only offers a high resolution of better than 10 ps but also exhibits strong flexibility, enabling easy reconfiguration for specific applications. Parameters, such as resolution, precision, and channel count, can be configured in both low and high modes. Subsequent studies have focused on discussing the addition or improvement of various fine-time interpolation modules, decoder modules, or calibration modules to enhance the measurement resolution and precision in the TDL-architecture TDC based on the WU method [8], [11], [26], [27], [28]. This section describes the subsequent modules. However, when the WU method is used, the pulse sequence generated by the WU launcher is longer, making the subsequent delay chain longer. It degrades the TDC performance owing to nonlinearity and timing jitter during the digital conversion process. Theoretically, the problem can be overcome by increasing the number of WU launchers, which results in extremely high resource consumption. Kwiatkowski and Szplet [29] proposed a new multisampling WU (MSWU) method to address the problems introduced by excessive sampling, such as long dead time and severe jitter. The method disregards the clock signal frequency and enables multiple snapshots of a WU to propagate along a delay chain. Deng and Chen [30] introduced a WU launcher into the STOP input signal using this method. Consequently, not only does the test signal generate multiple edges as the start signal enters the subsequent measurement modules, but the stop signal that triggers sampling also generates a pulse sequence with multiple edges before entering the delay chain. They implemented the TDC and achieved an rms resolution of better than 4.32 ps. In summary, the TDC signal input module that utilizes the WU method considerably improves measurement resolution and precision. However, further cooperation is required between the fine interpolation and encoding calibration modules to reduce the nonlinearity. In recent years, researchers have proposed new strategies to improve the WU method and explored using different hardware to implement it. Some scholars have attempted to use digital signal processing (DSP) blocks in FPGAs to realize the WU method [31], while others have implemented it on ASIC platforms [32]. These attempts and improvements have driven the progress of the WU method and the enhancement of the TDC performance. Accordingly, the WU method remains a popular research topic for the future.

In addition to adding a WU launcher to the signal input module to generate multiple edges for subsequent measurements by the TDC, the system clock or reference clock can be preprocessed before inputting it into the TDC to improve the signal-to-noise ratio (SNR) during subsequent signal sampling. In our previous study, we used an external high-precision clock generation circuit to produce a highly stable and low-noise reference clock signal, which was then input into the FPGA [33]. In addition to generating the STOP signal for the fine-time interpolation module, the signal was used as the sampling clock for other signals. The high-precision external reference clock provided strong assurance for the subsequent sampling and calibration of the other modules following the introduction into the FPGA via the signal input module. Overall, measurement results with a resolution below 20 ps were obtained using the on-chip high-precision system clock, and the robustness of multiple measurements was remarkably improved. However, the method introduces problems with internal and external clock synchronization and channel deviations may occur in multichannel scenarios. Therefore, solving the synchronization problem increases complexity in the subsequent coding modules and increases consumption of logic resources.

B. Coarse Counting Module

In addition to resolution and precision, dynamic range is another crucial performance parameter of an FPGA-based TDC. In theory, a single-frequency counter can be used to measure the time intervals. For example, if the counting clock frequency is 1000 MHz, it can achieve a resolution of 1 ns for time-interval measurements. Simultaneously, if the corresponding binary width of the counter is eight bits, it can realize time-interval measurements within the range of 1–256 ns. Therefore, it is clear that time-interval measurements with a simple structure can be performed using a simple counter. However, the problems they encountered were evident. First, the counting clock frequency must be increased to achieve a high resolution. However, the frequency cannot be infinitely increased at the technical implementation level, and the stability of a high-frequency clocks is poor. Once the frequency exceeds a certain threshold, it becomes unusable in FPGA devices and introduces significant noise and reflection phenomena in the ASIC. Second, even a slight increase in the clock frequency after overclocking results in unpredictable cost escalation, which is not desirable for an FPGA-based TDC. This simple structure is not without advantages. First, when researchers pursue ultrahigh resolution in the fine-time interpolation module, a simple counter can be used to extend the dynamic range by expanding the bit width of the counter. Therefore, many subsequent studies refer to this simple counter for a wide measurement range as a coarse counter [34], [35], [36], [37], [38], which corresponds to the coarse counter module in the entire TDC system. This classic architecture, which combines the coarse counter module with the fine-time interpolation module, has been successfully applied in almost every FPGA-based TDC. Second, the coarse counting method also provides a simple idea for subsequent research [39], [40], that is, how to achieve high-resolution time-interval measurements solely from the perspective of the sampling clock. They found that by utilizing on-chip or off-chip high-precision clocks combined with PLLs to generate multiple clocks of the same frequency but different phases, the limitations of a single- and ultrahigh-frequency clock could be overcome and time subdivision could be achieved. Subsequent studies shown that this fine interpolation method is practical with a high-performance architecture [41], [42], which is detailed in the fine-time interpolation module.

C. Fine-Time Interpolation Module

When attempting to perform precise time-interval measurements, a single-frequency counter is generally unable to complete the measurement when the resolution of the measured time interval is less than or equal to the nanosecond level. Therefore, an FPGA-based TDC relies on the fine-time interpolation module to obtain high-resolution measurements. Thus, the fine-time interpolation module is the critical component that determines the resolution, precision, nonlinearity, resource consumption, and power consumption of the entire TDC system. The literature review on FPGA-based TDCs reveals that the fine-time interpolation module has various architectures, each with different principles, structures, and implementation schemes, as well as different performance advantages, disadvantages, and applicable environments. In response to these architectures, a classification method was proposed [18] that initially resolved the ambiguity and lack of standardization in architecture classification, which were prevalent in previous research. To provide a simple, accurate, and standardized classification guide for fine-time interpolation architectures for future scholars entering the field of TDC, we have extensively and comprehensively organized previous extensive research and recent studies. Therefore, we propose a more concise classification method, as shown

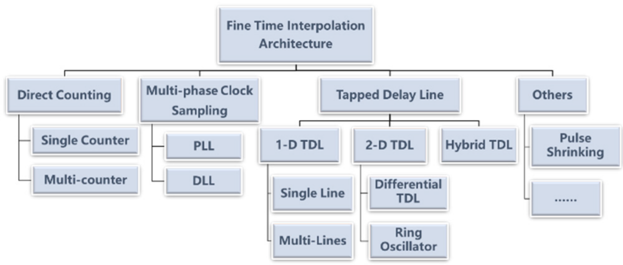

Fig. 3. Classification of fine-time interpolation architecture.

in Fig. 3. Based on our proposed classification method, fine interpolation methods can be divided into three categories at the top level of the topology: direct counting architecture, multiphase clock-sampling architecture, and TDL architecture. Subcategories can be gradually on the modifications made by different studies to the implementation details of fine-time interpolation methods. Fine interpolation methods other than these most common architectures, such as pulse-shrinking methods, are categorized as "others." These architectures are not mutually exclusive, and during specific implementations, one or more methods can be combined based on the specific application requirements to fully leverage the advantages of each architecture and compensate for its shortcomings.

1) Direct Counting Architecture: As stated in the coarse counting module, the simplest time-interval measurement solution in the TDC involves using a counter. In an FPGA, an on-chip system clock cycle is used to measure the time interval between one or multiple pulses. The counter is triggered directly or sampled by the signal to be measured. Thus, the architecture is also known as the direct counting architecture. To achieve high-resolution measurements using the direct counting architecture, a stable and high-frequency clock must be provided for the fine interpolation module. For example, a stable clock frequency of 100 GHz is required to achieve a resolution of 10 ps. However, currently, such a high-frequency clock cannot be stably implemented in an all-digital circuit, and prevailing manufacturing processes and technologies do not support the development of counters with such high flipping speeds. Therefore, a counter is generally used in a TDC system to complete coarse time measurements and can be used to expand the dynamic range of the measurements. Note that with the continuous improvement of technology and manufacturing processes, the direct counting architecture will still be the simplest and most effective means of precise time measurement if stable high-frequency clocks can be manufactured in the future.

2) Multiphase Clock-Sampling Architecture: In a TDC system implemented using an FPGA, the multiclock sampling method refers to sampling the data to be measured using multiple clocks created by a PLL or DLL. Generally, it uses multiple clocks with the same frequency but different phases to sample the data to be measured. During the gradual alignment of multiple clocks, the timestamp data obtained by decoding can be used to measure the time interval between the starting and ending pulses of the signal to be measured. The essence of this method entails using multiple phase-controlled sets of

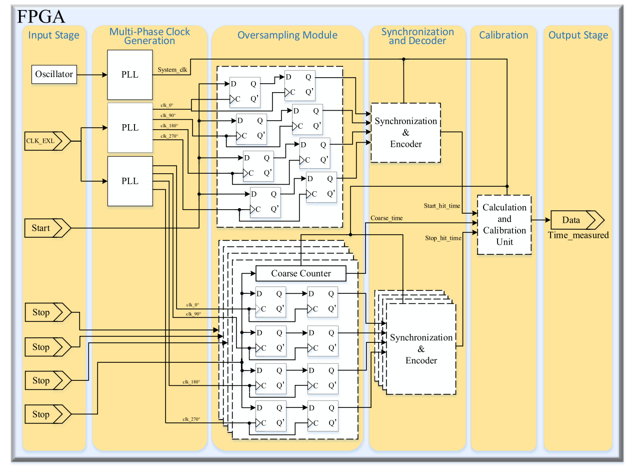

Fig. 4. Diagram of a typical multiphase clock-sampling architecture.

clocks to achieve a high-precision subdivision of the measured time interval and obtain high-resolution digital measurements of time signals based on an FPGA. Fig. 4 depicts a TDC implementation based on a typical multiphase clock-sampling architecture [44], [45]. The principle of this method involves using multiple clocks with the same frequency and constant phase difference to accurately locate the rising edges of the START and STOP signals. During the gradual alignment of the multiphase clocks, the fine time of the START and STOP ends can be calculated. Theoretically, the more same-frequency different-phase clocks we use, the better the time subdivision effect. However, this is limited by the physical structure of the PLL, which has a limited number of phase-shifted clocks that can be generated; the greater the number of clocks, the smaller the noise tolerance. The subdivision effect can also be improved by increasing the clock frequency; however, the method is limited by the onboard clock and the upper limit of the PLLs' frequency multiplication. Therefore, we must consider both factors—increasing the clock frequency and expanding the number of phases—to select the optimal clock frequency and the most reasonable number of phases. The ultimate goal is to achieve the highest time resolution.

Recently, an abundance of studies has examined the design and implementation of TDCs using multiphase clock-sampling architectures. In an application based on ToF methods, Spencer et al. [43] achieved subnanosecond resolution with a multiclock sampling TDC. They proposed two methods for implementing multiclock sampling in FPGA. The first method combines a single high-frequency driven counter (essentially a coarse counter) with four high-frequency driven phase-locked clocks (forming a fine-time division module) to digitize time. The second method involves directly using four independent high-frequency driven counters, each driven by a high-frequency clock with the same frequency and phase difference. The final time digitization was achieved by synchronously decoding the outputs of the four independent counters. The proposed methods will be validated in future studies. Wang et al. [44] implemented a 256-channel multiphase clock TDC and achieved a measurement resolution below 56.2 ps for dual channels. Their architecture followed the method described above. They demonstrated that the use of multiphase clock methods can lead to the development of high-precision TDCs with high resolution and suitability for high-performance applications. Using low-cost FPGA devices, Li et al. [45] implemented precise time measurements for avalanche photodiode detector systems using a multiphase clock-sampling TDC. They demonstrated the capability of the TDC to adapt to multichannel measurements with strong measurement stability. Even under poor incremental time input, it maintained a 1-ns full-width at half-maximum (FWHM) resolution and good linearity. Although a resolution of 1 ns is not considered high, it was evident from the results of the nuclear resonance scattering experiments that the TDC utilizing the multiphase clock-sampling method is suitable for performing multichannel measurements. Deng et al. [46] successfully applied a multiphase clock-sampling architecture to the TDC digital data acquisition module required in the prototype of a muon spin rotation, relaxation and resonance (μSR) spectrometer. To address the problem of high sensitivity to routing deviations caused by multiphase clocks, they employed solutions such as merging lookup tables (LUTs), reconfiguring the layout and routing, and utilizing zone constraints. Consequently, they reduced the routing deviation to 4 ps and achieved a high resolution of 184.7 ps. Moreover, the TDC featured 32 channels, satisfying the high-resolution and multichannel requirements of the μSR spectrometer prototype while consuming significantly less resources compared to other architectures. This approach effectively saves resources and area in the electronic readout module of the entire spectrometer.

By comparing and summarizing relevant research works [47], [48], [49], [50], [51], [52], [53], [54], the main advantages of TDCs based on multiphase clock-sampling architectures can be identified as follows: simple principles, capacity to achieve a resolution of 100 ps, low resource consumption, low power consumption, and strong stability and reliability in specific applications. However, several challenges exist in implementing multiphase clock sampling-based TDCs using FPGAs. First, it is difficult to ensure stable high-frequency clock sources. Many low-cost FPGA devices integrate multiple PLLs and onboard crystals; however, actual measurements have shown that the clock skew and jitter are high. In particular, when the clock frequency approaches the limit of PLL frequency doubling, it tends to exceed the noise tolerance, rendering the high-frequency clock output of the device unable to fulfill the requirements of multiphase clock sampling. Second, it is difficult to solve the high channel-to-channel skew of external clock sources. When the internal clock source is unstable, a high-frequency clock signal can be introduced from an external device as the reference or driving clock. However, even if an external clock is introduced via a subminiature version A connector (SMA) interface, a picosecond-level high channel-to-channel skew, caused by the different delays of clock signals in adjacent channels in the multiphase clock sampling method, remains. Consequently, it is difficult to improve the measurement resolution. Therefore, for both internal and external clock sources, if the tolerance limit for clock jitter cannot be fulfilled for a specific application, an external atomic clock should be used to improve clock jitter problems [55], [56]. Third, optimizing the layout and routing of clock signals is a complex task. Multiple PLLs are used when implementing the multiphase clock method, and the layout and routing of both the PLLs and input/output signals must be completed in the architecture design. A more compact layout and routing can reduce the signal propagation and onboard channel delays; however, the available onboard clock resources are limited when the number of channels and trigger frequency are extremely high. Therefore, optimizing the layout and routing of clock signals is crucial and complex. In summary, when selecting a multiphase clock-sampling architecture to implement a TDC, it is vital to comprehensively consider the performance requirements of the practical application and combine the design with the FPGA device to optimize the layout and routing of clock signals. This effectively reduces the impact of the aforementioned problems on the performance of the TDC system, enabling it to fully utilize the advantages of the multiphase clock-sampling architecture, which can achieve high resolution and reliability while saving logic resources.

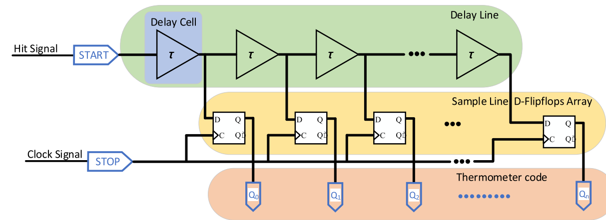

Fig. 5. Diagram of the 1-D TDL architecture.

3) TDL Architecture: The TDL architecture refers to a TDC architecture that utilizes the physical propagation delay of predefined logic resources in an FPGA device as the basic building blocks for implementing fine-time interpolation modules. Theoretically, the propagation delay of each logic unit in an FPGA is fixed and consistent. Therefore, as shown in Fig. 5, a propagation chain with taps can be constructed using delay elements with a fixed propagation delay. The input signal is fed into the propagation chain via a signal input module as the START signal, and it subsequently propagates along the chain. D flip-flops (FFs) are added after each delay element to sample the state output of each tap. When the edge of the STOP signal arrives, the D FFs output the sampling results. Based on the sampling results, the number of fixed delay units traversed by the input signal during propagation can be determined, and the measurement interval between the START and STOP edges can be obtained. A recent study proposed an improved TDL architecture based on the SuperWU method [20]. The least significant bit (LSB) measurement resolution was 366 fs, the single-shot rms precision is better than 12 ps, and the differential nonlinearity (DNL) and integral nonlinearity (INL) are as high as 250 fs and 2.5 ps, respectively. The dynamic range is extended to 10.3 s. Scholars have also packaged this architecture into an IP core for the ease of calling in specific applications. This work shows that TDC based on the TDL architecture can achieve subpicosecond or even femtosecond-level high resolution, but the fine-time interpolation module design must be tailored to the predefined resources of the selected FPGA, and the tradeoff between area occupancy and resolution should be fully considered during implementation. Generally, scholars refer to each delay unit and corresponding D FF on the constructed delay chain as a "bin" or "tap." Each bin completes the quantization task of converting the measurement interval into digital values in the TDL. Owing to the influences of the PVT, the propagation delay of each logic unit in an FPGA is inconsistent, which results in an uneven size distribution of bins. Therefore, to satisfy the specific requirements for the resolution, precision, and linearity of the TDL-TDC in different applications, the TDC architecture based on a conventional TDL must be improved to expand the measurement range, compensate for nonlinear errors, and improve the measurement resolution.

Owing to the widespread application and research of TDC based on the TDL architecture, solutions have been proposed for the aforementioned problems, which can be summarized into three categories. A classic Nutt interpolation method with the addition of coarse counters was employed to address the problem of expanding the measurement range. Second, nonlinear errors can be compensated by improving the subsequent decoder module by adding calibration modules, which are discussed in the following. Third, the measurement resolution can be enhanced by adding and improving a subinterpolation module, which is the sub-TDL fine-time interpolation module of the architecture. Various implementation methods exist for sub-TDL modules. For example, multiple TDLs can be created, and the results can be averaged. A hybrid TDL can be constructed by combining other fine interpolation methods, a 2-D TDL can be developed, or a TDL based on a ring oscillator can be utilized to obtain solutions tailored to the requirements of specific applications.

To improve the measurement precision, multiple studies have proposed using a fine interpolation method that increases the number of TDLs. It involves measuring the same signal using multiple sets of TDLs to form equivalent coding rows, averaging the results, and outputting them. Theoretically, the measurement precision of this fine interpolation module can be proportionally improved by increasing the number of TDLs. For example, Zieliński et al. [57] achieved a measurement precision of 500 ps using two TDLs, increasing the measurement precision from the nanosecond to 100-ps scale. In 2017, a multichain TDL architecture was implemented in high-end FPGA devices, resulting in an ultrahigh rms resolution of 3.9 ps [58]. They demonstrated that multiple TDLs not only effectively improve measurement precision but also have the potential to be designed and implemented in high-end FPGA devices. Another study evaluated the performance of single-, dual-, and quad-chain TDL architectures on the same FPGA device, achieving measurement results with an rms resolution greater than 10 ps [59]. To investigate whether the multichain TDL architecture can achieve high precision on low-cost FPGA devices, Wang et al. [60] used a multichain TDL architecture based on classical Nutt interpolation on a Xilinx Artix-7 FPGA, achieving the best rms resolution of 6.4 ps without introducing dead time. For a single TDL, differences in the bin width can lead to uneven quantization of the measured time intervals, particularly when only one measurement exists, resulting in a significant decrease in measurement precision. Although it may initially seem that using multiple TDLs can improve the measurement precision through equivalent repeated measurements for averaging effects, actual measurements have demonstrated that the quantity of average values finds it difficult to compensate for in ultrawide bins. This implies that as long as ultrawide bins exist, even when multiple TDLs are used, the measurement precision will not significantly improve because the ultrawide bins still dominate the averaged measurements. Therefore, in future research on TDL architectures with multiple TDLs, it will be necessary to consider the problem of ultrawide bins caused by the propagation of signals across the logic array blocks. An effective approach is to use the WU method, as described in Section II-A, to effectively subdivide ultrawide bins.

In addition to the averaging effect achieved by increasing the number of replicated TDLs, another classical approach involves constructing a fine-time interpolation architecture using two TDLs with a time difference, as depicted in Fig. 6. Kalisz et al. [61] proposed the use of a differential operating mode to improve the fine interpolation effect of TDL architectures to enhance the measurement resolution of TDCs. They achieved an excellent resolution of 200 ps within a measurement range of 10 ns, while most TDCs based on TDL architectures could only achieve a subnanosecond measurement resolution. The differential operating mode refers to a TDL architecture composed of two delay chains with different propagation speeds that sample the input signals and perform time-interval measurements. Owing to the similarity in principle with a Vernier caliper, the method is also referred to as the Vernier TDL in many studies [62], [63], [64]. This Vernier-like differential operating mode can also be implemented using two ring oscillators operating at slightly different frequencies. Therefore, the fine-time interpolation module based on ring oscillators can be classified as a differential TDL architecture. In summary, the differential TDL utilizes the STOP signal to catch up with the START signal in the fast- and slow-delay chains with a delay difference. The triggering edges of the START and STOP signals are detected using FFs. The measurement is completed when the STOP signal catches up with the START signal. The measured time interval amplified by the Vernier caliper principle can be calculated using the subsequent decoders based on the intersecting positions of the two signals. Based on the assumption that the STOP signal catches up with the START signal after passing through N delay elements, the digital conversion quantization formula for the measured time interval in a differential TDC architecture is given as follows:

$$T = N \cdot T_{\mathrm{LSB}} \tag{1}$$

where T_{LSB} represents the resolution of the differential TDL architecture, and its expression is

$$T_{\mathrm{LSB}} = \mathrm{Delay}_1 - \mathrm{Delay}_2 \tag{2}$$

where Delay_1 and Delay_2 represent the delay unit time of the slow and fast TDLs, respectively, which correspond to τ_1 and τ_2 shown in Fig. 6. If a ring oscillator is used to implement

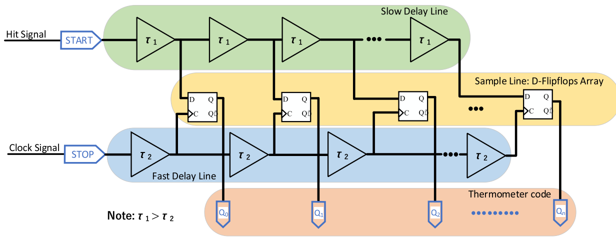

Fig. 6. Diagram of the differential TDL architecture.

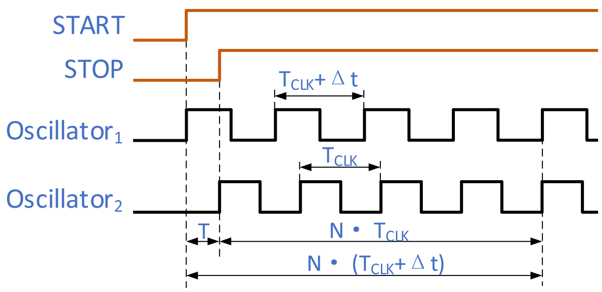

Fig. 7. Timing diagram of the Vernier method using two ring oscillators.

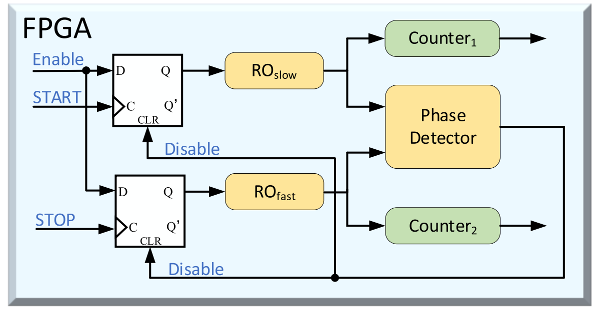

Fig. 8. Schematic of the Vernier interpolation method.

the differential architecture [65], [66], [67], [68], [69], [70], [71], [72], [73], [74], it is known that two slightly different frequencies of the ring oscillators exist, as shown in the timing diagram of Fig. 7, with their corresponding periods represented as T_{CLK} and $T_{\mathrm{CLK}} + \Delta t$. The digital conversion quantization formula for the measured time interval in this case is given as

$$T = N \cdot (T_{\mathrm{CLK}}+\Delta t) - N \cdot T_{\mathrm{CLK}} = N\cdot\Delta t \tag{3}$$

where Δt signifies the period difference between the two ring oscillators and N indicates the number of clock cycles elapsed when the rising edge of the fast oscillator coincides with and catches up to the rising edge of the slow oscillator, as detected by the phase detector.

According to the above principles, the Vernier interpolation method is primarily implemented using a fast- and slow-ring oscillator waveform generator, phase detector, and counter, as depicted in Fig. 8, where separate counters are used for the two oscillator loops to enhance the measurement rate. Analogous to (1) and (3), it can be observed that the TDLs of the differential architecture, whether built using delay elements or ring oscillators, are based on the Vernier principle. The resolution of the Vernier caliper is equal to the minimum scale difference between the main and Vernier scales. By applying this principle to the fine-time interpolation module of the TDC, the resolution of the TDC was determined by the time difference between the two TDLs or the period difference between the two ring oscillators. Therefore, the TDLs of the differential architecture can effectively enhance the measurement resolution of the entire TDC by reducing the time difference between the TDLs or making the frequencies of the ring oscillators closer. A low-complexity multichannel ring oscillator architecture TDC proposed by Berrima et al. [75] used only 53 FFs and 29 LUTs resources in an FPGA device to achieve a single-shot precision of 92.9 ps. They primarily adopted the fine-tuning delay (FTD) routing method to balance the impact of routing delays of START and STOP signals on the overall precision of the system. Another study improved the traditional ring oscillator architecture into a TDC with three operating modes: multichannel, high precision, and high throughput [76]. The worst rms precision of the proposed TDC in these three operating modes is 34 ps, providing excellent performance for specific application environments and saving time in balancing the indicators. However, the architecture exhibits certain limitations. First, catching up between fast and slow TDLs requires a long time, which increases the dead time of the measurement. Second, the technology requires extremely high clock frequency stability and minimizes the phase difference of the clock period, which increases both the implementation difficulty of the TDC and the cost.

To avoid the long dead time generated by the traditional TDL architecture and improve the pulse-to-pulse resolution of the TDC system, researchers have proposed a 2-D TDL called a delay matrix as shown in Fig. 9, which arranges the delay units that constitute a traditional single TDL into a rectangular array [77]. Two delay matrices run in parallel with one delay matrix used to capture the events that occur at the current time, while the other completes the processing of events captured by the previous event. Each row of the delay matrix can be considered a single TDL, and the rows are vertically connected only through the delay units in the first column and share the same input clock. The experimental results have shown that using this architecture, the TDC resolution can reach 150 ps, and the resolution for the dual-pulse signal input is 2.5 ns. In terms of performance, it achieved substantial improvements compared to the single TDL architecture at that time. This 2-D TDL enables the continuous detection and processing of the events under test through two parallel running delay matrices. This working mode effectively reduces the dead time but also brings about problems such as duplicate counting of the same pulse by multiple-row TDLs and metastability problems when cascading registers. Therefore, Amiri et al. [78] proposed a reverse reset scheme to eliminate erroneous readings of the tapped values. Moreover, they specifically considered the sensitivity of the delay matrix size when implementing it on FPGA devices and designed a set of criteria. Through testing, a TDC based on a 2-D TDL architecture achieved a measurement resolution of 100 ps on a low-cost FPGA device. Chen et al. [79] proposed a PLL-based delay matrix to address the inability to fully control the delay values of each delay unit in the ordinary 2-D TDL architecture. In their proposed architecture, all the delay units are accurately controlled by

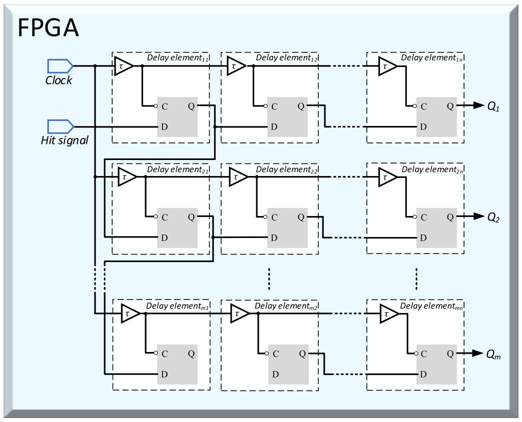

Fig. 9. Schematic of the 2-D TDL architecture.

the PLL to achieve high-precision phase division and generate uniform output phases. Experimental data show that the architecture can achieve overclocking and double the data rate, obtaining a high resolution of 15.6 ps. In another recent study, a TDC based on a cross-TDL architecture was employed to achieve time-to-space conversion in a high-resolution imaging instrument [80]. The architecture is essentially a 2-D TDL architecture. This application-oriented TDC was implemented on programmable logic and achieved an rms resolution of 15 ps and a detection rate of 10 Mcps, demonstrating that the architecture not only has high performance in experimental measurements but also has strong practical significance. However, implementing a 2-D TDL architecture requires careful consideration of the delay unit control in the delay matrix. The use of a DLL and PLL can solve the problem and ensure the uniformity of the output phases, ultimately eliminating the influence of the PVT on the TDC performance and achieving a deterministic 2-D delay matrix.

Increasing the number of TDLs can improve the resolution owing to the averaging effect. Moreover, changing the delay time of the TDLs or utilizing the frequency difference in ring oscillators to construct differential TDLs and 2-D TDLs can also achieve good performance. However, these TDL architectures have drawbacks, such as increased dead time, increased cost, and high consumption of logic and routing resources. Consequently, they are not allowed in many application scenarios where resource consumption, power consumption, dead time, and cost are sensitive factors. Therefore, it is crucial to study how to improve the fine-time interpolation method or the interpolation module structure to enhance the resolution and precision using only a single TDL.

Aloisio et al. [81] proposed the use of TDLs as the second stage and a digital clock manager (DCM) phase shifting method as the first step to realize fine interpolation. The TDLs used in the second step are selected using a state machine. The method essentially drives the TDLs with DCM phase shifting, resulting in the generation of four equivalent TDLs. They deployed the method on low-cost FPGA devices and achieved a measurement resolution higher than 60 ps. Through analysis, we determine that this method is a hybrid TDL architecture that combines single TDLs and multiphase clock-sampling methods. The architecture effectively leverages the advantages of the two architectures and overcomes their respective shortcomings, thereby achieving high measurement resolution and precision. Based on this idea, Chen and Li [82] integrated four methods into a TDL architecture and successfully implemented 96 channels in two types of FPGA devices. In addition to achieving a resolution better than 10 ps and a nonlinearity of 0.01 LSB, only 25% of the logic resources were consumed. Another study improved the measurement precision by improving the sampling mode of the delay chain [83]. The output of the same delay chain is sampled twice using the two edges of the sampling clock; hence, it is also called a dual-sampling architecture. The architecture was implemented and compared on Cyclone and Spartan 3 E FPGA devices, both of which achieved a stable rms precision of more than 100 ps. Multiple studies have proven that reconfiguring delay units and forming new delay-chain schemes with large propagation delays can effectively reduce the nonlinearity errors caused by uneven bin widths of delay units with fixed delays. Among them, Chen [84] proposed a merged TDL that reduced DNL by 16.6% and INL by 5.4% and achieved a measurement resolution better than 50 ps and an rms precision of 29 ps. Subsequently, Chen [85] proposed a counting weighted delay line (CWD) for the PET scanners. In their architecture, a counting weighting method was specifically designed for the delay times of the delay units. The DNL calculated using this method is significantly lower than that calculated using the code density method. Moreover, the architecture eliminates the need for nonlinear calibration and saves a considerable number of resources and power. The concept of a pseudo-segmented TDL has been previously proposed [86]. The pseudo-segmented TDL connects multiple D FFs to each tap to achieve the same measurement resolution as that of the multi-TDL architecture. Therefore, the architecture is referred to as equivalent coding TDL in other studies [87]. Deng et al. [88] proposed a nonuniform monotonic multiphase (NUMMP) method to construct TDLs that overcome the inherent propagation delay of FPGA logic units. The basic principle of this architecture is to reassemble delay units with nonuniform delay times and combine them with time scale marking methods, feature extraction, and decoding algorithms to achieve parameter-adjustable high-resolution TDC. Experimental results have shown two modes of this architecture on 28-nm FPGA: one mode with low nonlinearity, namely, 20-ps LSB resolution and [−0.05, +0.06] LSB DNL, as well as [−0.15, +0.08] LSB INL; the other mode achieved a resolution as high as 1.87-ps LSB at the expense of linearity performance.

Several recent studies have provided improved ideas for single TDL architectures. Hua and Chitnis [89] designed and implemented a novel uncompensated single-TDL architecture. The architecture differs from traditional thermometer code architectures in that it encodes the TDL state. The experimental results at resolutions ranging from 5 to 87.74 ps exhibited excellent linearity, all within 1 LSB. In addition, the resource consumption of this architecture is only half that of traditional TDL architectures, and the interleaved histogram method ensures continuous time-to-digital conversion without dead time. In addition to improving TDLs constructed using fixed delay logic units, Wang et al. [90] have used LUTs as delay elements to construct TDLs. To demonstrate the potential of this TDL architecture, they designed and implemented a TDC on three of the most advanced FPGA chips: Xilinx Virtex-7, Ultrascale, and Ultrascale+. The results showed excellent measurement resolution and linearity for all three chips. The architecture uses a novel sampling matrix method to improve the resolution and utilizes virtual bin calibration methods for automatic calibration. Using LUTs as the basic delay unit results in extremely low resource consumption. In addition to achieving good performance, the novel approach also provides a strong reference for future improvements in TDL architectures.

As mentioned above, the fine-time interpolation module based on the TDL architecture primarily focuses on design improvements at the level of the delay elements used in constructing the TDLs, the sampling methods for each output stage of the TDLs, and the configuration of the delay chain. Improvements in the implementation step primarily involve targeted logic resource allocation, clock distribution, layout, and routing for the predefined resources of FPGA chips. We believe that future iterations and optimizations of the fine-time interpolation module will continue to follow this research direction.

D. Decoding Module

The decoder module in the TDC is responsible for decoding and outputting the binary code generated by the fine-time interpolation module. This module is also referred to as an encoder in various studies [91], [92]. As the output code of the module reflects the binary code of precise time, it requires a specialized decoder design for the three different architectures of the fine-time interpolation module and its implementation on an FPGA chip. The direct counting architecture TDC generally acts as a coarse counting module in the TDC system because the period of the counting clock determines the measurement resolution; however, the clock frequency cannot be infinitely increased. Therefore, it is unnecessary to discuss the decoder problem in this architecture. In the multiphase clock-sampling TDC architecture, multiple clocks with the same frequency and a fixed phase difference perform fine-time interpolation. Along with the sampling logic and edge detection logic, each parallel measurement line ultimately generates “0” or “1” output code, as shown in Fig. 4. This code reflects the starting and ending positions of the fine-time jump to be measured. Accordingly, the decoder module in this architecture converts the output code of each parallel measurement line into a specified number of binary codes. The decoder module plays an extremely crucial role in the widely used TDL architecture of the TDC. Its task is to convert the thermometer code output into a specified number of binary codes for the output using the fine-time module. Generally, improvements are made to the fine-time interpolation and sampling logic to improve the measurement resolution and precision of the TDC with the TDL architecture. Therefore, the decoder module must be designed and implemented accordingly.

Multiple studies have been dedicated to improving the design of decoder modules and conducting experiments on TDC systems [7], [8], [10], [21], [58], [71], [90]. The experimental results indicate that improvements in the decoder module can effectively enhance the precision and reduce the nonlinearity of the TDC while also reducing resource consumption. In each measurement process of fine time, deviations in the trigger setup and hold time, clock signal uniformity skew, and uneven propagation on delay lines can result in bubbles (“0” surrounded by “1” or “1” surrounded by “0”) in the sampled output codes. These bubbles can lead to erroneous conversions during decoding and increase system errors. Therefore, specialized anti-bubble decoding strategies must be developed. Research has indicated two approaches for achieving anti-bubble decoding [93]. First, only the propagation bits in the sampled output of the fine-time interpolation are counted. The quantity of “1” in the count thermometer code is used to directly calculate the binary code. This method consumes fewer logical resources and theoretically ignores the bubble problems. Second, a dedicated decoding logic is designed to identify the moment of occurrence of the first propagating bit in the sampled output code. The solution can accurately detect the positions of signal transitions and prevent incorrect decoding of bubble bits. However, the approach requires significant computational resources. Subsequently, targeted improvements were implemented to address the limitations of the anti-bubble decoding strategies. For example, one study referred to anti-bubble solutions in the design of ADCs [94]. In terms of decoding principles, they employed a binary search method. It involves finding the location of “1–0” transitions in the thermometer code and repeats the process to obtain the output binary code. For the on-chip implementation, they utilized LUT primitives and place and route (PAR) constraints in Xilinx FPGA chips to improve the speed and stability of the decoding module. Two recent studies also improved the anti-bubble decoding schemes and achieved favorable performance. In a resource-saving TDL architecture, Kong et al. [95] divided the sampled output subsequence into two parts: “0–1” transitions and “1–0” transitions, which were separately decoded using subdecoders. The binary code was then directly constructed from the results of the two subdecoders, and the outputs of all the subdecoders were summed to obtain a binary code representing the fine timestamp. The decoding scheme can be easily implemented by utilizing the abundant LUT resources in an FPGA. In another study, an efficient predecoder was designed for the TDC using the WU method [96]. The decomposition and clustering software techniques were partially hardware-based and implemented on an FPGA. The experimental results demonstrate that the predecoder can effectively limit bubble errors, reduce logic complexity during decoding with the WU method, and improve the conversion efficiency.

All the above decoding schemes can be categorized as thermometer-to-binary (T2B) logic, which is easy to implement on an FPGA, but reduces the throughput and increases the dead time of the TDC system. To address this problem, Dutton et al. [97] have proposed using XOR-based multievent position encoding as an alternative to T2B logic. The outputs sampled from the triggers are directly connected to the XOR logic, and the output of the XOR logic is fed into a synchronous ripple counter to create a histogram. This decoding scheme essentially encodes the positions of multiple events in the sampled output codes, effectively improving the conversion rate and reducing dead time. In another study, a priority decoding algorithm was employed to calculate the measured time of a deserialized signal [98]. The decoding scheme can reduce the FPGA chip area utilization. In a TDC system with dual-channel cascaded delay-line architecture, a pipelined decoding module was applied [99]. The module employs a pipelined design to accurately detect the positions of multiple edges on delay lines and can precisely identify six different waveforms. Experimental results reveal that the decoding scheme can achieve a low dead time of two clock cycles and a measurement resolution better than 9 ps. In summary, the current research on TDC decoding modules primarily focuses on reducing the logical complexity, improving conversion rates, minimizing dead time, and reducing resource consumption and area utilization.

E. Calibration Module

As described in the fine-time interpolation module, different interpolation architectures use different time-to-digital conversion principles and their implementation on different FPGA devices is limited by predefined resources. In summary, the TDC linearity is affected by the FPGA manufacturing process, operating temperature, and power supply voltage (PVT). The impact primarily manifests as process variations that can cause uneven propagation delays when constructing delay lines based on FPGA-fixed delay units, significantly reducing the linearity of the TDC. Clock skew can degrade the measurement precision of TDCs based on multiphase clock-sampling architectures, and an increase in operating temperature and power supply voltage jitter can also deteriorate the TDC linearity performance. These scenarios are commonly referred to as introducing nonlinear errors to TDCs, which seriously degrade the measurement precision. The specific linearity characterization method is detailed in the performance parameters section of Section III. Here, we primarily discuss calibration methods that can reduce the nonlinearity of TDC measurements.

Nonlinear calibration methods have been used to reduce the nonlinear errors of the TDC system and improve the measurement precision. Calibration techniques primarily focus on two aspects: 1) the problem of zero-width bins or over-width bins caused by uneven propagation delay or clock skew and 2) the nonlinearity problem caused by increased operating temperature and power supply voltage fluctuation. To address the first problem, studies have proposed combining multiphase clock sampling with a delay-line architecture to reduce the nonlinearity introduced by clock skew [100], [101]. The experimental results showed that the standard deviations of DNL and INL were reduced to 0.01 and 0.02 LSB, respectively, after using the bin-width calibration method, with LSB being 10.5 ps. Most studies have adopted a bin-by-bin calibration method, also known as bin readjustment [102], [103], [104], [105]. It involves creating a table based on the bin-size data obtained from code density testing, which are stored in the on-chip memory of the FPGA. The data in the table can remain unchanged or be updated in real time based on the PVT conditions, ultimately achieving real-time online calibration. This bin-width calibration method directly intervenes and adjusts the size of the bins to make them more uniform, ultimately reducing quantization errors and nonlinearity. The method plays a crucial role when the propagation delay along the delay-line unit is highly uneven and only requires minimal resources for porting and environmental adaptation to different TDC architectures and FPGA devices. Xie et al. [106] proposed a mixed-binning (MB) calibration method that can achieve adjustable resolution without increasing the nonlinearity. The bin-width calibration method used in the MB calibration is implemented based on block random access memory (BRAM), making it more suitable for multichannel applications than bin-by-bin calibration using a large number of LUTs or block RAMs. Moreover, it has been suggested that the bin decimation method can be used as an alternative to bin-by-bin calibration to minimize nonlinear errors [107]. It is a bin-width calibration method whose principle is to group physical delay units with a fixed propagation delay and construct larger bins. Upon regrouping, the bin size becomes uniform, and the improvement in the delay unit uniformity directly increases the linearity of the TDC based on the delay-line architecture. When implemented on an FPGA, the method does not increase resource consumption. However, experimental results have revealed that it can reduce measurement resolution to some extent. Therefore, when choosing this method, it is necessary to balance resolution, precision, and hardware resource consumption. To address the second problem, Szymanowski and Kalisz [108] have used DLLs to indirectly stabilize the uniformity of the delay elements. These DLLs can resist temperature variations and can be used to control the voltage of the FPGA. By designing and implementing these loops, researchers achieved a measurement uncertainty of better than 140 ps and a nonlinearity of approximately 0.1 LSB. Wang et al. [109] applied a temperature compensation method to compensate for temperature variations. The experimental results showed effective nonlinear compensation within the temperature range of 30 °C–60 °C, and good linearity was achieved. Based on it, Rogacki and Zurbuchen [110] designed and implemented an automatic compensation calibration module that compensated for both temperature and voltage variations, successfully reducing the standard deviation of the nonlinearity from 60 to 15 ps. In a more recent study, a temperature compensation method was used to detect the peak power of nanosecond pulses in a TDC operating across a wide temperature range [111]. The calibration method utilizes the temperature characteristics of the delay lines built inside an FPGA to effectively address the problem of inaccurate energy measurements caused by temperature drift. In summary, a nonlinear calibration module is indispensable for the TDC. Currently, research on the design and implementation of calibration methods has primarily focused on improving the uniformity of delay elements, reducing the influence of clock skew, resisting temperature and voltage variations, implementing real-time online calibration, and reducing resource consumption. In addition to the improvements that can be made at the design and implementation levels, it is necessary to recognize that most calibration methods utilized in research still require manual calibration, which does not fulfill the increasing demand for automatic calibration in various applications. Therefore, Choi and Jee [112] and Wang et al. [113] have proposed strategies such as gain and weighted calibrations to achieve automatic calibration. However, these methods cannot flexibly configure resolution. Although some studies have achieved adjustable resolution, their means of reconfiguration are manual [88], [89]. Collectively, these problems indicate that future research on calibration methods must focus on simultaneously achieving automatic calibration and online resolution configurations.

By analyzing and sorting various modules involved in implementing TDC on FPGA chips, we have found that the fine-time interpolation module is crucial to affecting the TDC performance. Most studies have improved the measurement resolution or precision of TDC by improving the fine-time interpolation architecture. Therefore, we have classified and introduced these architectures in detail, summarized their advantages and disadvantages, and identified their most suitable applications. For example, although the resolution achieved by the direct counting architecture is limited by the system clock frequency, it is widely used for coarse time counting to expand the measurement range of TDCs. The multiphase clock-sampling architecture, with a resolution generally only in the nanosecond or 100-ps range, has a simple principle and extremely low resource consumption, making it suitable for application systems with high generality requirements and sensitivity to resource consumption. The TDL architecture, as the most extensively studied architecture in recent years, can achieve the highest measurement resolution among several architectures. Studies on improving its precision and linear performance are also constantly emerging. Therefore, it can fully leverage its advantages in applications such as high-energy physics experiments and ToF mass spectrometers, which require extremely high measurement resolution. However, this architecture requires high FPGA hardware performance and complex decoding and calibration algorithms, which can considerably increase resource consumption, power consumption, and costs. In brief, it is necessary to determine the performance requirements of the TDC based on the application requirements and select the appropriate FPGA device for TDC implementation. Once the hardware is selected, the most suitable TDC architecture should be designed and implemented for specific application.

III. Performance and Evaluation

Upon summarizing the architecture and implementation methods of FPGA-based TDC technology, it is necessary to assess and compare the overall performance of TDCs. The performance of FPGA-based TDCs can be characterized by analyzing several key parameters that affect the performance of the TDC. These performance metrics are universal and must be evaluated in future TDC designs and implementations. Different FPGA devices and implementation architectures yield different performance results; therefore, it is essential to perform both horizontal comparisons of these performance metrics and vertical analysis of their developmental trends.

A. Resolution

Generally, the measurement resolution of the TDC refers to the minimum time interval that the TDC can distinguish. It is also the smallest increment in the time-to-digital conversion curve, which is the minimum quantization step of the input–output transfer characteristic curve of the TDC. FPGA-based TDCs use the LSB to characterize their resolution performance, with smaller values indicating a higher resolution. As TDCs are widely used in fields related to ToF, an increasing number of studies have focused on improving their resolution performance, with resolution enhancement being one of the primary measures of progress [114], [115], [116], [117], [118], [119]. However, different FPGA devices and architectures can lead to different measurement resolutions. From a hardware perspective, more advanced manufacturing processes for FPGA devices are more likely to achieve higher resolution. Considering the implementation architecture, the resolution of a direct counting architecture is equal to that of the system clock cycle. To enhance the resolution of this architecture, only stable and high-frequency clock signals, such as *LC* oscillators, can be used; however, the use of high-frequency clock signals is limited by FPGA chips and cannot be infinitely improved, which increases the power consumption and cost of the entire TDC system. Therefore, the direct counting architecture is suitable for low-resolution applications (greater than 2 ns).

The resolution of a multiphase clock-sampling architecture is based on the number of sampling clocks; theoretically, the more sampling clocks with the same frequency and fixed phase shift, the higher the measurement resolution of the TDC. However, the number of PLLs capable of generating multiple sampling clocks in an FPGA is limited, and it is considerably difficult to layout and route across clock domains. Therefore, the multiphase clock-sampling architecture is suitable for medium-resolution applications that range from hundreds of picoseconds to 1 ns. Although the TDL architecture has a basic delay-line structure and multiple improved variants [120], [121], [122], [123], their essence is the same. The resolution of the architecture depends on the fixed physical delay of the basic delay unit used to construct the delay line. The smaller the delay value of the delay unit, the higher the resolution of the entire delay line. Theoretically, advanced FPGA devices can achieve fixed unit delays of less than 100 ps. However, if the logic units or dedicated carry logic of a specific FPGA device are used as the basic delay unit for constructing a delay line, the delay value of the delay unit cannot be changed at the hardware level. Therefore, intensive research has been conducted to optimize the delay-line creation methods, making delay values uniform, and reducing the impact of environmental factors [124], [125], [126], [127]. By adopting these improvement strategies, the resolution of the TDC based on the TDL architecture can be increased to less than 10 ps or even subpicosecond levels. This resolution is currently the highest level achievable for fabricating FPGA-based TDCs. Therefore, a TDC based on the TDL architecture is suitable for high-resolution applications, and using improved architectures, such as multichain delay lines, Vernier delay lines, and ring oscillators, can further enhance the resolution performance.

B. Precision

Generally, the measurement precision includes two aspects: accuracy and precision. Accuracy refers to the influence of systematic errors, whereas precision refers to the influence of random errors. The measurement precision of the TDC refers to the difference between the measured and expected values of the TDC system, which is typically measured using the standard deviation and is also referred to as single-shot precision. In theory, the precision of the TDC follows a Gaussian distribution. However, actual measurements have shown that the distribution curve of the TDC measurement precision does not always exhibit an ideal distribution. To quantitatively describe the measurement precision of a TDC, scholars have generally input test signals with fixed time intervals into the TDC system and evaluated the measurement precision by calculating the standard deviation of the time-interval measurement results using the following formula:

$$\sigma^2 = \frac{1}{N-1} \cdot \sum_{1}^{n} (x_i - \mu)^2 \tag{4}$$

where x_i represents the bin width of the ith time-interval measurement result and μ indicates the average value obtained from N measurement results. Clearly, this precision evaluation method is based on the hardware measurement results of the time intervals and utilizes the average value for calculation and assessment. However, this may either overestimate or underestimate the precision level of the TDC, failing to accurately reflect its actual precision. Therefore, research has found that by performing time-interval tests to measure the standard deviation of the quantization error and jitter and calculating the rms value of each standard deviation using (5) [128], [129], [130], the value represents the precision of the TDC, which is also known as its rms resolution

$$\sigma_{\mathrm{TDC}} = \sqrt{\sigma_s^2 + \sigma_q^2} \tag{5}$$

where σ_{TDC} represents the rms value of the TDC, σ_s signifies the signal drift error generated during the propagation of the test signal, and σ_q denotes the time-to-digital conversion quantization error of the TDC. These values were calculated using the following equations:

$$\sigma_q = \frac{\mathrm{LSB}}{\sqrt{6}} \tag{6}$$

$$\sigma_s = \sqrt{\sigma_{\mathrm{CLK}}^2 + \sigma_{\mathrm{INL}}^2 + \sigma_{\mathrm{in}}^2 + \sigma_L^2} \tag{7}$$

where LSB represents the least significant bit of the TDC measurement result, which is equivalent to the resolution value. σ_{CLK} indicates the skew of the system clock, σ_{INL} represents the cumulative drift during the propagation of the test signal within the TDC architecture, which is generally expressed by the standard deviation of the INL. σ_{in} indicates the drift of the input signal itself, and σ_L denotes the circuit drift of the input stage within the architecture. The parameter usually refers to the deviation introduced by the cross-clock domain circuit connection in a multiphase clock-sampling architecture or the interline drift in a multidelay-line architecture. Therefore, σ_L can be ignored in a single delay-line architecture.

Using the aforementioned methods and equations, the measurement precision of the TDC can be evaluated using time-interval testing. The evaluation results are also referred to as rms resolutions in some studies. Therefore, although the resolution index has been divided into LSB and rms in certain studies [70], the rms value represents the measurement precision. Research on improving the TDC measurement precision can be conducted from two aspects: reducing the quantization error and drift error. As the quantization error essentially characterizes the measurement resolution, improving it can reduce the quantization error. However, to enhance measurement precision, it is necessary to reduce the drift error by considering both the internal and external aspects of the TDC architecture. Yu et al. [33], Choi and Jee [112], and Chen [131] have developed dedicated layouts and routing for different architectures and FPGA devices to minimize the internal drift while leveraging the chosen FPGA hardware resources. The clock skew is reduced by introducing external high-precision, low-jitter clock sources, and the input signal drift is mitigated by designing specialized noise suppression circuits. The cumulative drift of the test signal during propagation within the architecture can be reduced by implementing nonlinear calibration, thereby diminishing the INL. These drift suppression approaches must be considered and implemented in each specific TDC design to achieve higher measurement precision.

C. Measurement Range

The measurement range represents the time interval in which the TDC can measure. Since the time interval is scalar, the measurement range of the TDC is generally expressed as the maximum value of the time interval that can be measured before a system overflow occurs. The design, implementation, and related applications of TDCs require a continuous expansion of the dynamic measurement range. For example, millisecond-level measurement ranges may be required in certain laser ranging applications. For the three TDC implementation architectures described in Section II-B, if only the fine-time interpolation module is used to achieve a larger measurement range, it would consume a significant amount of FPGA chip area and logic resources, which is not feasible in most applications. Therefore, regardless of the TDC architecture used, the measurement range was expanded by introducing a dedicated coarse counter module. The coarse counter module can easily extend the counting range of the entire period by increasing the number of counter bits, thereby expanding the measurement range of the entire TDC. Furthermore, owing to the simple implementation principle of the coarse counter and its low resource consumption, the combination of a coarse counter and a fine-time interpolation module, also known as Nutt interpolation, has become the typical and most commonly used architecture in reported FPGA-based TDC implementations [132], [133].

D. Nonlinearity

Linearity is a crucial indicator of the static characteristics of the TDC. It refers to whether the output of the TDC system can maintain a linear relationship with the input, similar to that of an ideal system. In a TDC, the nonlinearity error is a commonly used metric to characterize linearity performance. It is primarily caused by variations in the chip manufacturing process, operating voltage and temperature (PVT), uneven delay time of the delay elements in the fine-time interpolation module, and signal crosstalk problems that occur when fabricating the FPGA board. The nonlinearity error is defined as the deviation between the actual transfer curve of the TDC and ideal quantization fitting line. The nonlinearity error in the TDC was measured using DNL and INL. DNL refers to the difference between the actual step size and the theoretical step size in the input–output transfer characteristic curve of the TDC system. INL represents the nonlinearity error in a single time measurement of the TDC, which is generally represented by the cumulative DNL. As DNL can be expressed in terms of the minimum quantization step, also known as LSB, both the DNL and INL values are expressed in units of one LSB time. During the measurement process, code density tests are commonly employed to measure the DNL and INL values, and histogram plots are used to visually represent the measurement results. For example, in a TDC implemented based on a delay-line architecture with a fine-time interpolation module, the DNL represents the deviation between the measured delay time of the actual delay element and the ideal delay time or its average value. It reflects the variation in the delay time of the delay element around the ideal value. Its numerical value can be calculated using (8). The INL represents the nonlinearity of the entire delay chain, and its value can be obtained by integrating the DNL, as shown in (9)

$$\mathrm{DNL}_i = t_i - \bar{t} \tag{8}$$

$$\mathrm{INL}_i = \sum_{i=0}^{N} \mathrm{DNL}_i = \sum_{i=0}^{N} \left(t_i - \bar{t}\right) \tag{9}$$

where DNL_i represents the DNL value of the ith delay element, INL_i represents the INL value of the ith delay element, t_i indicates the measured delay time value of the ith delay element, and $\bar{t}$ indicates the theoretical delay value of the delay element calculated as the average value from multiple measurements. Theoretically, the delay time for each delay element is fixed and consistent. In the code density test method, t_i and $\bar{t}$ are calculated using the following equations:

$$t_i = n_i \cdot \frac{T_{\mathrm{CLK}}}{N} \tag{10}$$

$$\bar{t} = \frac{T_{\mathrm{CLK}}}{N'} \tag{11}$$

where N indicates the number of measurements performed by the TDC, T_{CLK} denotes the system clock period, n_i represents the count of the ith delay element recorded during the code density test, and N' signifies the number of delay elements. To normalize the units of DNL and INL, it is common to divide the values obtained from the equations for DNL_i and DNL_i by $\bar{t}$. Accordingly, the results are unified in terms of LSB as the unit.

Considering the research on TDC based on the TDL architecture as an example, the nonlinearity of the TDC is primarily caused by the inconsistency between the delay time of each delay element and the theoretical delay time, that is, the uneven distribution. Furthermore, this unevenness depends on the structure of the TDC, and the layout and routing of the FPGA. Therefore, extensive research has shown that designing a TDC with a reasonable structure and adjusting the layout and routing within and between modules can effectively improve the nonlinearity [134], [135], [136], [137]. In addition, reducing the length of the delay chain can also decrease the nonlinearity; however, it limits the measurement range of fine-time interpolation modules based on the delay chain. Therefore, when designing and implementing a TDC based on the TDL architecture, a tradeoff should exist between the measurement range and the nonlinearity performance of the TDC. In conclusion, whether based on TDL or other architectures, reducing nonlinear errors is a crucial factor to consider. Therefore, specifically designed nonlinear calibration modules were used to address this problem [138], as elaborated in detail in Section II-E.

E. Dead Time

The dead time refers to the preparation time required for a TDC system to complete the interval measurement of the input under test and restart the measurement upon detecting a change in the input signal. Moreover, it is the minimum time interval that the TDC system must pass after the input signal changes. The dead time is essential for the TDC to accurately detect and measure changes in the input signal. During the dead time, changes in the input signal were ignored to avoid errors and uncertainties. Therefore, the longer the dead time, the slower the measurement speed and conversion efficiency of the TDC. Consequently, many studies have been devoted to designing and implementing methods to reduce the dead time [44], [78], [126], [128]. Fan et al. [139] performed simultaneous measurements of time and pulsewidth using only one channel in time-over-threshold measurements. The experimental results indicated that the dead time is approximately two clock cycles. A shorter dead time enables the TDC to operate effectively in high event rate applications in high-energy physics experiments and improves the speed of the measurement system. In another study aimed at a measurement rate of 60 MSamples/s [37], strict control and experimental verification of dead time were conducted, enabling the single-photon avalanche diode (SPAD) measurement system to measure and establish histograms within a few seconds.

The existence of dead time is caused by signal propagation delays and the stability requirements in digital circuits. Once a measurement is completed and the input signal changes, a certain amount of time is required to ensure signal stability and allow the TDC system to accurately capture the new input state. The dead time can be a fixed time-interval set in the hardware design or a delay time set in software algorithms. The dead time depends on specific applications and design requirements. Although a shorter dead time can improve the response speed and conversion efficiency of a TDC system, it can also increase the effects of noise and instability. However, a longer dead time can reduce errors and interference, but the response speed of the TDC for high-speed application scenarios may be insufficient. Therefore, when designing and implementing FPGA-based TDCs, it is crucial to balance the dead time and system conversion efficiency based on specific application requirements. The goal is to minimize the dead time while ensuring accurate measurements using the TDC system.

F. Resource and Power Consumption

The design and implementation of a TDC on an FPGA platform are different from those of an ASIC. Once the specific model of the FPGA chip is determined, its resources are predefined in advance, and the designer only needs to study how to effectively utilize the on-chip logic resources for TDC architecture design, as well as how to layout and route them based on on-chip resource positioning. It is not a concern in the ASIC-based TDC design because all resources on the ASIC can be customized. On the FPGA platform, as all the modules of the TDC system are based on predefined on-chip resources, ensuring the implementation of a multichannel TDC and reserving sufficient logic resources for other logical modules in specific applications have become the focus of research [90], [95], [99], [119]. In summary, it is crucial to consider resource optimization when designing the TDC to fully utilize the limited logic and storage resources in an FPGA and reduce resource consumption. The main methods include the following.

1) Optimizing the architecture of the TDC implementation, primarily by improving the design of the fine-time interpolation module, reducing the complexity of the decoder module, and making reasonable improvements to the calibration module. Optimizing the design of these main modules of the TDC system can significantly reduce the overall resource consumption of the TDC.
2) When choosing FPGA devices, selecting low-power devices as much as possible based on application requirements can reduce resource consumption. Low-power devices also help reduce resource consumption.
3) Optimizing the FPGA hardware structure, such as adopting effective pipelined architectures and eliminating redundant calculations, can effectively reduce resource consumption.
4) Partial duplication can be used to implement the TDC. The method can effectively reduce the resource consumption by properly allocating resources and dynamically loading the code. In conclusion, FPGA-based TDCs must be balanced and optimized based on specific requirements and resource limitations in different application scenarios to improve the performance and efficiency of the converter.

Reducing resource consumption helps save the FPGA area and lower TDC costs while reducing power consumption in the overall system. The power consumption in a TDC refers to the energy consumed during digital conversion. As a TDC generally comprises digital circuits and clock signals, the digital circuitry must perform switching, calculations, and processing operations among the electronic components, resulting in energy loss and conversion. For an FPGA-based TDC implementation, the power consumption can be estimated by analyzing the on-chip current consumption or voltage drop. The magnitude of the power consumption depends on various factors, including the TDC design, circuit complexity, operating frequency, and power supply voltage. Generally, a higher power consumption indicates that greater energy is expended when the device operates. For low-power TDC devices, designers have focused on optimizing circuit structures, selecting low-power devices and algorithms, controlling the clock tree depth, and shortening signal transmission paths. Implementing energy-saving measures is crucial for reducing power consumption. In addition to enhancing the reliability and efficiency of the TDC system, the efforts also extend its lifespan, ultimately reducing overall costs. Therefore, power consumption serves as a crucial metric for assessing the TDC energy efficiency and effectiveness. Designers strive to minimize power consumption while fulfilling the performance requirements to improve the energy efficiency and sustainability of the device.

Regarding the performance metrics mentioned above, designing and implementing an FPGA-based TDC requires careful consideration and a tradeoff between the measurement resolution, precision, system linearity, dead time, resource consumption, and power consumption. It is crucial to evaluate the interplay and constraints among these factors to select the most suitable FPGA device, TDC implementation architecture, and performance assessment approach.

IV. DISCUSSION

Based on a summary and analysis of the implementation architecture and performance evaluation of FPGA-based TDC, significant advancements and progress have been achieved in recent years. Table I comprehensively summarizes the capabilities of FPGA-based TDCs in terms of resolution, precision, measurement range, linearity, resource consumption, and power consumption. This table compares and analyzes the parameters to identify the existing constraints in the current FPGA-based TDC and to provide areas for improvement and enhancement in future research. Ultimately, it can be used to explore new emerging applications in which FPGA-based TDCs may find potential suitability.

As presented in Table I, we classify and summarize the literature from recent years based on various methods for implementing the fine-time interpolation module in the TDC. Research on TDCs based on the TDL architecture remains the most prevalent because, as FPGA technology continues to advance, the available logic resources for constructing delay lines with low delay times become more abundant. These resources include a dedicated carry logic with shorter delay times, lower jitter, adaptive logic modules (ALMs), and a richer set of LUTs. However, the basic delay units used in constructing the delay lines are affected by the PVT factors of the FPGA, resulting in uneven delay times that severely affect the measurement precision and linearity of the TDC. Therefore, a majority of the research has focused on exploring methods to suppress the effects of PVT and improve the uniformity of the bin-width distribution, rather than proposing new alternative solutions. Detailed explanations

TABLE I
OVERVIEW AND COMPARISON OF RECENT FPGA-BASED TDCS' PERFORMANCE

Ref.-Year	**FPGA Device**	**LSB**	**Precision**	**Measurement Range**	**DNL**	**INL**	**Dead Time**	**Power Consumption**	**Resources Consumption**
Unit		ps	ps	ns	LSB	LSB	ns	mW	
Multi-phase clock sampling architecture									
[41]-2016	Kintex-7	89.3	56.2	—	[0.44,0.87]	[0.44,0.82]	4.3	—	—
[42]-2017	Spartan-6	1000	—	4096×10^3	—	—	32	—	—
[51]-2017	Kintex-7	280	80-100	37×10^3	±0.4	<0.05	6.25	4100	4361LUTs
[152]-2017	Stratix IV	2.5	6.72	16.5	[-0.56,0.46]	[-2.98,3.23]	19.75	212	35675LEs
[43]-2019	Virtex-6	184.7	—	327.68×10^3	<0.044	<0.044	<10	—	—
[39]-2021	Kintex-7	250	—	1344	±0.25	[-0.46,1.47]	—	—	—
Tapped delay line architecture									
1-D TDL TDCs									
[77]-2018	Virtex-7	10.54	14.59	—	[0.05,0.08]	[-0.09,0.11]	—	—	712Slices
[78]-2018	UltraScale	5.02	7.8	—	[-0.12,0.11]	[-0.15,0.48]	—	—	80CARRYs
[25]-2019	Artix-7	2	9.7	157×10^3	—	2.1	110	<13.8	7010LUTs
[28]-2019	Altera-SoC	38	15	—	±0.58	—	2.5	—	364ALMs
[100]-2019	Cyclone-V	5.8	6.6	—	[-0.98,3.25]	[-10.85,11.69]	—	1100	—
[118]-2019	Kintex-7	3.6	5.3	20	—	—	2.86	—	646LUTs
[124]-2019	UltraScale	305×10^{-3}	8.5	10.24×10^3	—	—	—	—	3200Slices
[125]-2019	UltraScale	10.2	18	—	±2	±4	—	—	184CARRYs
[82]-2020	Kintex-7	1.11	<5.3	383×10^6	[-0.98,3.73]	[-17.47,38.56]	—	—	300Slices
[90]-2020	Kintex-7	<20	9.7	20	—	—	—	390	836LUTs
[107]-2020	Artix-7	4.88	2.9-8.03	350×10^3	[-0.10,0.15]	[-0.23,0.28]	—	—	2962LUTs
[126]-2020	Virtex-5	46.875	35.5	6	[-0.78,0.76]	[-7.2,4.3]	—	—	6391LUTs
[132]-2020	Virtex-7	105.73	—	853	±0.02	[-0.06,0.03]	—	—	986Slices
[132]-2020	UltraScale	50.92	—	853	[-0.03,0.06]	[-0.02,0.09]	—	—	356Slices
[17]-2021	Artix-7	0.366	12	10.3×10^9	<250fs	<2.5ps	5	20.2	7010LUTs
[20]-2021	UltraScale	1.23	3.67	—	[-0.84,7.93]	[-6.36,24.7]	—	1030	88CARRYs
[55]-2021	Artix-7	14.89	6.4	2.467×10^6	—	[-10.37,27.46]	—	—	6193LUTs
[83]-2021	Kintex-7	20	7.7	—	[-0.05,0.06]	[-0.15,0.08]	8	435	828LUTs
[83]-2021	Kintex-7	1.87	2.2	—	[-0.54,1.30]	[-2.21,3.51]	8	740	1679LUTs
[94]-2021	Virtex-6	9	6.2	—	[-0.9,3.66]	[-4.27,26.01]	5	—	765LUTs
[133]-2021	Kintex-7	15	—	1000	<4ps	<10ps	—	3.89	—
[140]-2021	Zynq-7000	9.83	13.86	—	[-0.14,0.16]	[-0.25,0.42]	—	—	764LUTs
[35]-2022	Artix-7	18	<54	1×10^6	[-0.95,2.37]	[-0.78,1.63]	20	—	348LUTs
[84]-2022	UltraScale+	5	—	5	[-0.99,1.44]	[-2.84,1.62]	0	—	—
[86]-2022	Artix-7	22.1	22.35	262.14×10^3	[-0.71,1.05] [-0.73,1.06]	[-0.85,0.86] [-1.17,0.04]	4	164	228LUTs
[101]-2022	UltraScale	51.28	15.89	1000	[-0.018,0.021]	[-0.019,0.035]	—	—	9472CARRYs
[122]-2022	Cyclone-10	5.68	4.8	100	[-0.96,2.90]	[-9.98,11.14]	8.3	—	—
[144]-2022	Artix-7	10	17	164×10^3	[-0.13,0.15]	[-2.26,3.54]	<60	—	21591LUTs
[145]-2022	Spartan-6	72.4	39.3	20×10^3	[-0.56,0.56]	[-0.86,0.76]	495	12.12	19852LUTs
[11]-2023	UltraScale+	0.46	<9	100	[-1,6.84]	[-2.57,72.55]	—	—	234CARRYs
[30]-2023	Artix-7	17.9	<20	—	1.006	0.920	5	130	1212LUTs
[85]-2023	UltraScale+	4.2	5.9	41.9×10^6	[-0.6,0.2]	[-0.2,2.3]	7.5	—	458CARRYs
[91]-2023	Kintex-7	0.4	<5.2	500×10^3	[-0.97,5.95]	[-8.02,219.3]	—	—	811LUTs
[147]-2023	Kintex-7	6.5	6.4	50	[-1,4]	±15	—	—	231LUTs
2-D TDL TDCs									
[59]-2020	Virtex-6	10	5.5	5	[-0.07,0.07]	[-0.13,0.03]	—	—	—
[74]-2020	Stratix-IV	15.6	15.6	1000	[-0.157,0.137]	[-0.176,0.184]	—	—	1126LUTs
[71]-2021	Kintex-7	43	34 14 33	45	[-0.98,0.08]	[-1,0.97]	50 50 6.25	—	—
[75]-2022	Artix-7	2.17	<12	145×10^3	—	<4ps	7	—	—
[85]-2022	UltraScale+ UltraScale Virtex-7	20.97 36.01 34.84	17.11 27.37 32.33	—	[-0.055,0.034] [-0.036,0.046] [-0.033,0.034]	[-0.196,0.000] [-0.057,0.081] [-0.016,0.277]	—	—	455LUTs 453LUTs 437LUTs
Other architectures									
[139]-2018	Actel Flash	8.5	<42.4	10	[-0.36,0.36]	[-0.91,0.91]	—	—	36BUFDs
[28]-2022	UltraScale+	2.38	<5.66	0.21	[-0.23,0.22]	[-0.24,0.27]	—	5	10Slices

regarding this matter can be found in Section II, specifically in the delay-line section of the fine-time interpolation module.

Thus far, the main architectures that significantly differ from the delay-line architecture include the multiphase clock-sampling architecture and the architecture based on pulse shrinking. The multiphase clock-sampling architecture is particularly attractive for applications where a high measurement resolution is not critical (generally in the nanosecond range) or for applications that involve multiple measurement channels, owing to its simple principles and low resource consumption. However, the pulse-shrinking method has limited application in FPGAs owing to the difficulty in controlling the shrinking factor.

Although a small number of studies have reported research on TDC architectures based on the pulse-shrinking method [140], [141], [142], [143], [144], the conclusion drawn is that while the pulse-shrinking method provides a new approach for achieving high-performance TDC on ASIC platforms, it is unsuitable for FPGA platforms. We believe that the selected FPGA device and external inputs, such as high-frequency clocks, also influence the overall performance of the TDC. Therefore, it is necessary to strictly use the method of controlling variables to conduct comparative studies on different architectures to better illustrate the problems and identify methods to improve and enhance the same performance parameters for different architectures as well as different performance parameters for the same architecture. Therefore, it is necessary to strictly use the method of controlling variables to conduct comparative studies on different architectures to better illustrate the problems and identify how to improve and enhance the same performance parameters for different architectures, as well as different performance parameters for the same architecture.

Once the FPGA device is tailored for specific applications and the intended TDC fine-time interpolation architecture is determined, enhancing the measurement resolution and precision of the FPGA-based TDC becomes the focus of research. We summarize the efforts made in the input, sampling, and data-decoding output modules of the TDC system by reviewing the methods employed in related studies. Incorporating a WU launcher into the input module is an effective means for achieving subpicosecond resolution potential. Adjusting the sampling method during the sampling stage can also enhance the measurement precision. The core idea involves performing multiple repeated measurements on a single measurement channel and then average them or increase the number of measurement channels to achieve an averaging effect. Research on improvement methods for data-decoding and output modules is diverse. The primary focus is on improving the encoding and decoding strategies, as well as adding dedicated calibration modules for error compensation and correction. All these improvement measures are based on the selected FPGA device and specified application.

Although the TDC performance will be improved at the hardware level with the continuous advancement of FPGA technology, such as in more recent research that combines high-performance FPGAs with high-speed data processing hardware, such as ARM or DSP, to enhance the overall performance of TDC systems [31], [113], [145], [146]. It primarily utilizes the high-speed parallel processing capability of high-performance FPGAs and the powerful computational capability of ARM or DSP. These approaches are potential research directions for improving TDC research as hardware performance improves and will also provide new ideas for the emergence of new architectures. Therefore, further research on this topic is necessary. However, more important and valuable TDC research should be application oriented. In early years, TDC was primarily applied to high-energy physics experiments, and most related research reports were published based on this application. With the increasing demand for applications based on ToF in recent years, most TDCs have been specifically developed for ToF measurement systems. For example, Deng et al. [46] developed a 32-channel multiphase clock-sampling architecture TDC module with low dead-time requirements for μSR spectrometers. Although the achieved measurement resolution was only slightly better than 184.7 ps, they solved the severe routing skew problem in the multiphase clock-sampling method, reducing the routing skew of all channels to less than 4 ps. It significantly improves the reliability of TDC in μSR spectrometers, and the use of Ethernet interface protocol in the data transmission scheme facilitates large-scale data transfer when expanding channels in the future. Hua et al. [147] implemented a ToF pulse measurement with a 45-ps measurement resolution TDC using a high-performance UltraScale+ FPGA chip and integrated it with commercial-grade SPAD detectors and low-cost small-sized laser sources into a pulse oximeter. The study demonstrated the great potential of TDC in emerging applications, such as wearable devices. In addition, in low-rate deep-space optical communication application [148], a TDC is used to digitize the output of superconducting nanowire single-photon detectors. The measurement results show that using a TDC can significantly reduce the analog-to-digital throughput, effectively improving the transmission rate of downstream data. In addition to validating the performance of the TDC, these application studies have also demonstrated the significant advantages of developing high-performance, low-cost, and easily portable precision time measurement systems based on FPGA platforms.

With the continuous application of FPGA-based TDCs in the high-speed acquisition cards of scientific instruments, such as high-performance spectrometers and mass spectrometers, increasingly strict constraints exist on power consumption and resource usage. However, many methods capable of effectively reducing the nonlinearity of TDC measurements and improving the measurement accuracy, such as expanding the number of measurement channels, increasing the number of delay lines, and adding calibration modules, will lead to increased resource and power consumption. Therefore, many scholars have begun exploring methods to achieve low power and resource consumption in low-cost FPGAs. For example, Xie et al. [136] combined the TDC with an SPAD array for measurement and proposed an MB calibration method to improve the linearity of the TDC and reduce resource consumption. Moreover, they conducted comparative experiments on two different manufacturing processes of FPGA platforms to verify that low-cost hardware can achieve high performance and low resource consumption. Portaluppi et al. [149] validated a TDL architecture TDC based on the WU-A method for time-correlated single-photon counting. The experimental results demonstrated the correctness of the proposed continuous code density calibration scheme and laid the foundation for splitting and decoding when expanding the number of channels to reduce hardware resource consumption. Another study implemented a low-power and low-resource-consuming 33-channel TDC and achieved a resolution of approximately 72.4 ps using a reasonable calibration strategy.

Such low resource consumption enables implementation of a 128-channel TDC on low-cost FPGA devices [150]. Based on the above analysis, we can foresee that future research on improving the TDC performance will no longer solely focus on improving the resolution and precision while ignoring the hardware resource consumption and overall power consumption of application systems.

Based on the above summary and analysis, we found that FPGA chip technology is constantly advancing, providing an increasing number of resources for TDC research. However, the current and future research trends for TDC are shifting toward applications rather than experimentation. Furthermore, the pursuit of system reliability for specific applications continues to increase. Problems that could be disregarded in a laboratory environment will be magnified. Therefore, based on the analysis of existing research achievements on FPGA-based TDC, we summarize the following urgent problems and potential challenges that need to be addressed.

1) Hardware conditions for achieving high-performance TDCs continue to improve, and research needs to keep up with the times by proposing new architectures and methods. Several years ago, Xilinx released the Ultrascale series FPGA chips manufactured using a 20-nm process. These chips offer the following advantages. First, more advanced dedicated carry logic provides ultrashort delay and low drift physical delay units for delay-line architectures. Second, more stable clock resources, including high stability, low-jitter on-chip clocks, and PLLs, can provide more clock signal outputs, facilitating higher resolution time interpolation for multiphase clock-sampling architectures. Third, richer LUT and storage resources have increased the logic resource capacity and on-chip storage space, even with a reduced chip area, enabling the expansion of multichannel TDCs. Finally, a more rational resource layout and abundant routing resources not only reduce layout and routing difficulties but also make the layout more compact, effectively reducing routing deviations. In fact, numerous studies have proposed high-performance TDC designs using the new chip [14], [82], [90], [129], [130], [147], [151]. While this is just one concrete example of FPGA chip technology advancement, it reflects the need to combine new hardware and propose new fine-grained time interpolation architectures or innovative methods for input, sampling, decoding, and calibration modules that can fully utilize these resources to improve TDC measurement resolution.
2) Fine-time interpolation architecture is crucial for achieving high resolution and precision in TDCs. Currently, research on this architecture is focused on maximizing the channel count of low-cost FPGA devices. However, because of the limited resources provided by low-cost hardware, it is necessary to improve the fine-time interpolation architecture to adapt to the selected hardware and broaden the application market. More recently, several improvement strategies have been developed for the TDL architecture. One strategy involves reducing the length of the delay line and utilizing new encoding schemes [95]. The objective is to maintain the resolution and precision while enhancing the linearity and resource efficiency. Another approach involves using folded delay lines to interpolate the system clock cycle [152]. This method not only achieves an rms precision better than 10 ps but also reduces resource consumption by over 50% compared to conventional delay-line architectures. A cyclic approach has also been explored, such as organizing delay lines in the form of Vernier-based ring oscillators [70]. This approach effectively folds the delay lines into shorter ring oscillators to reduce mismatches and maintain uniformity to improve the linearity. Another strategy focuses on maximizing the utilization of delay units to save a significant amount of logic resources. Kwiatkowski and Szplet [87] proposed an architecture based on pseudo-segmented delay lines and chopped time coding lines. They achieved an average resolution of 1 ps and measurement precision of more than 4 ps on a low-cost Kintex-7 chip. Furthermore, the concept of a multiphase clock-sampling architecture is inherently resource-efficient. Although this architecture cannot achieve the same resolution as the TDL architecture, its strong potential for channel expansion, simple implementation, and convenient portability makes it more feasible in specific applications [42], [46]. Future research trends of this architecture will mainly focus on reducing the difficulty of manual layout and routing, the nonlinearity caused by routing deviation, and the sampling synchronization problem under multiple high-frequency clock drives. In addition to improving the aforementioned architectures individually, a hybrid architecture combines and leverages the advantages of different architectures in a cascading manner to achieve high performance [72], [153]. However, it is an approach that generally requires substantial resource consumption and necessitates the consideration of synchronization problems among the different stages of fine-time interpolation. This does not align with the development trends of high integration and low resource consumption in TDCs.
3) Owing to severe bubble problems and measurement nonlinearity caused by PVT factors, high-performance TDCs must implement nonlinear calibration. The research focus has been on improving the linearity while minimizing resource consumption of the calibration module. One approach involves using an online calibration module that stores a calibration histogram on a chip that can be updated in real time. This method is often referred to as bin-by-bin calibration [154]. In recent years, improvements have been made to this method. For example, Chen and Li [82] implemented a direct compensation architecture and hybrid calibration method that did not consume additional resources while removing bubbles and zero-width bins. They achieved DNL and INL performances better than 0.1 LSB and an LSB resolution better than 10 ps on both 20- and 28-nm FPGAs. The improvement was achieved with highly linear performance without sacrificing

measurement resolution. Several calibration schemes have been derived from this calibration method, such as bin decimation, MB calibration, virtual bin calibration, and other methods that effectively overcome bubble errors, improve linearity, and reduce resource consumption [90], [106], [155]. However, these methods still rely on manual calibration methods, such as density testing of temperature codes, which may result in the loss of time information during calibration. Although automatic calibration methods have been proposed and implemented, such as gain and offset calibration and weighted calibration, these methods can only achieve a fixed resolution and are unsuitable for applications that require a flexible configuration of the TDC resolution. Hua and Chitnis [89] proposed a novel TDC that encodes the state of the delay line and uses interleaved histograms to maintain continuous conversion without introducing dead time. This provides a novel method of state-based encoding for single delay lines rather than the traditional temperature-based encoding. The TDC does not require compensation, thus fundamentally solving the problem of nonlinear calibration and providing a new direction for future calibration module designs. This suggests that the cumbersome and complex calibration process can be replaced with more reasonable sampling and encoding/decoding strategies. In conclusion, the research trend in calibration strategies aims to improve the design for specific applications, thereby enabling online resolution configuration and automatic calibration.

4) The dead time of a TDC determines the amount of time it takes to prepare for the next measurement after completing one; with the continuous development of precision time measurement applications, TDCs must strive for shorter dead times to achieve faster sampling rates. The dead time varied among the TDCs using different time interpolation architectures. By comparing different studies, it was found that TDCs based on the TDL architecture had shorter dead times. This is because the differential delay lines constructed based on the Vernier principle using ring oscillators with slightly different frequencies to achieve fine-time subdivision can lower the working frequency of the ring oscillators to the lowest allowable value, thereby reducing dead time. However, the method increases power consumption and is subject to the limitations of the FPGA chip in terms of the changing frequency. Another method for reducing dead time is to construct a 2-D time delay matrix [79]. Compared to regular 1-D linear delay lines, this delay line constructed using a 2-D structure can effectively improve the measurement resolution of each pulse and reduce dead time; however, it increases resource consumption. In summary, the demand for high sampling rates in specific applications will continue to drive research toward reducing the dead time to almost zero.

5) The evaluation of the measurement uncertainty of an FPGA-based TDC is a crucial task that requires further research. As a precision time-interval measurement device, the precision of the TDC is influenced by various factors, such as the time-to-digital quantization process, nonlinearity of the fine-time interpolation module, timing jitter caused during the layout and routing of signals, inherent drift and noise of the measured signal, and reference clock skew. Hence, evaluating the measurement uncertainty is necessary to ensure the reliability of TDC measurement results and improve measurement precision. Szplet et al. [156] analyzed the aforementioned sources of errors affecting TDC precision and performed an uncertainty evaluation on a two-stage interpolation architecture TDC. Their work provided constructive recommendations for enhancing the accuracy and precision of the TDC based on specific mathematical models and derived formulas for calculating the total uncertainty. Their findings indicated the necessity of an uncertainty evaluation for all types of TDCs. However, our investigation revealed a lack of research on measurement uncertainty evaluation, specifically for FPGA-based TDC, except for the analysis conducted by Xie et al. [23] on the single-stage TDC that they developed. Therefore, future research should include the evaluation of the measurement uncertainty after completing the design, implementation, and experimental verification of the TDC. It is a crucial and meaningful task because it holds significant guiding value for performance assessment, quality control, and improvement of the measurement process in specific application environments of FPGA-based TDCs.

6) As mentioned earlier, current research on TDCs is shifting toward improving the design and implementation methods for specific applications, focusing on low costs, short development cycles, and easy hardware portability to fulfill market demands. This requires the designed TDC to have strong portability to satisfy different application requirements. Specifically, it should be capable of rapid architectural reconstruction, rapid resetting of the calibration parameters, and high integration. By reviewing the recent literature on TDC applications [157], [158], [159], [160], [161], [162], [163], [164], [165], we believe that each TDC module should be integrated into an IP core with an external operating interface for customer use. These demands have driven competition and elimination in TDC architectures. For example, many TDCs aimed at practical applications adopt multiphase clock-sampling architectures instead of the more commonly reported TDL architectures. This is because the nonlinear calibration problems of TDL architectures require redesigning with specific hardware, which significantly increases the difficulty of their application. However, many TDC application studies have addressed this problem by implementing sample data output and calibration using microprocessors, such as ARMs or DSPs [118], [145]. Improving the decoding and calibration strategies for microprocessors is relatively easy, enabling the integration of TDC modules based on the FPGA design and providing data interfaces

to microprocessors. In addition to the overall integration of the TDC systems mentioned above, currently reported research results on TDCs lack a comprehensive summary and comparative analysis of the calibration mechanisms. This is primarily because most calibration schemes must be considered in conjunction with specific hardware or applications, which makes it difficult to evaluate them uniformly. However, in future research, it will be crucial to summarize and analyze the performance of existing calibration strategies in terms of linearity, resistance to PVT effects, robustness, and portability. It is essential for researchers to select an appropriate calibration strategy for specific applications.

In reviewing the literature on FPGA-based TDCs, we estimate that TDCs will be driven by application markets to innovate and develop in the future. The performance of TDCs in specific applications is correlated with progress in FPGA chips. Advanced manufacturing processes for FPGA devices must be used to improve measurement resolution, precision, and linearity. However, reducing the resource and power consumption also needs to be considered, as they are related to the total cost of the TDCs. Therefore, we must comprehensively weigh and consider various performance indicators in conjunction with the specific application requirements. Overall, TDCs will gradually move out of the laboratory and will be widely used in commercial precision time measurement equipment. Commercial TDCs will require high integration and portability, necessitating further investments in the development of automated TDC development tools. Only in this manner can we continuously narrow the gap between TDCs developed on FPGAs and TDCs developed on ASIC platforms, reduce development costs, and expand their application in the market.

V. Conclusion

This article presented an analysis of FPGA-based TDC architecture and implementation methods organized in the order of various modules in the TDC system design and implementation. A standardized and concise classification method was proposed for the TDC fine-time interpolation architecture to unify the classification of architectures with various names. In addition, we reviewed the relevant literature on TDC technology and analyzed performance indicators, such as resolution, precision, measurement range, linearity, resource consumption, and power consumption, achievable by TDC technology. We identified the technical problems faced using different modules during TDC system implementation, including the limitations of different fine-time interpolation architectures and suggested potential solutions. Furthermore, we analyzed the performance of TDCs in different application scenarios to determine the strengths and weaknesses of the different architectures. Finally, we compared and analyzed the measurement results obtained in recent studies and provided a reference in terms of experience summary, development trends, and potential applications for future research on FPGA-based TDC technology through a comprehensive review of the published literature.

References

[1] M. Ikeno et al., "A FASTBUS TDC system," *IEEE Trans. Nucl. Sci.*, vol. NS-33, no. 1, pp. 779–782, Jan. 1986, doi: 10.1109/tns.1986.4337214.

[2] H. Ohgaki, H. Ijiri, T. Sakae, N. Koori, and M. Matoba, "A ratio-to-digital converter for charge division position sensing," *IEEE Trans. Nucl. Sci.*, vol. NS-34, no. 1, pp. 110–114, Mar. 1987, doi: 10.1109/TNS.1987.4337311.

[3] R. Bassini, C. Boiano, S. Brambilla, C. Carriero, and M. Malatesta, "A 32 channel TDC on a VME board," *IEEE Trans. Nucl. Sci.*, vol. 45, no. 3, pp. 858–863, Jun. 1998, doi: 10.1109/23.682651.

[4] J. Wu and Z. Shi, "The 10-ps wave union TDC: Improving FPGA TDC resolution beyond its cell delay," in *Proc. IEEE Nucl. Sci. Symp. Conf. Rec.*, Oct. 2008, pp. 3440–3446, doi: 10.1109/NSSMIC.2008.4775079.

[5] J. Wang, S. Liu, L. Zhao, X. Hu, and Q. An, "The 10-ps multitime measurements averaging TDC implemented in an FPGA," *IEEE Trans. Nucl. Sci.*, vol. 58, no. 4, pp. 2011–2018, Aug. 2011, doi: 10.1109/TNS.2011.2158551.

[6] H. Menninga, C. Favi, M. W. Fishburn, and E. Charbon, "A multi-channel, 10 ps resolution, FPGA-based TDC with 300 MS/s throughput for open-source PET applications," in *Proc. IEEE Nucl. Sci. Symp. Conf. Rec.*, Oct. 2011, pp. 1515–1522, doi: 10.1109/NSSMIC.2011.6154362.

[7] R. Szplet, D. Sondej, and G. Grzeda, "Interpolating time counter with multi-edge coding," in *Proc. Joint Eur. Freq. Time Forum Int. Freq. Control Symp. (EFTF/IFC)*, Jul. 2013, pp. 321–324, doi: 10.1109/EFTF-IFC.2013.6702285.

[8] Q. Shen et al., "A fast improved fat tree encoder for wave union TDC in an FPGA," *Chin. Phys. C*, vol. 37, no. 10, Oct. 2013, Art. no. 106102, doi: 10.1088/1674-1137/37/10/106102.

[9] C. Liu and Y. Wang, "A 128-channel, 710 M samples/second, and less than 10 ps RMS resolution time-to-digital converter implemented in a Kintex-7 FPGA," *IEEE Trans. Nucl. Sci.*, vol. 62, no. 3, pp. 773–783, Jun. 2015.

[10] R. Szplet, P. Kwiatkowski, Z. Jachna, and K. Rózyc, "An eight-channel 4.5-ps precision timestamps-based time interval counter in FPGA chip," *IEEE Trans. Instrum. Meas.*, vol. 65, no. 9, pp. 2088–2100, Sep. 2016, doi: 10.1109/TIM.2016.2564038.

[11] Y. Wang and C. Liu, "A 4.2 ps time-interval RMS resolution time-to-digital converter using a bin decimation method in an ultrascale FPGA," *IEEE Trans. Nucl. Sci.*, vol. 63, no. 5, pp. 2632–2638, Oct. 2016, doi: 10.1109/TNS.2016.2606627.

[12] Y. Wang and C. Liu, "A 3.9 ps time-interval RMS precision time-to-digital converter using a dual-sampling method in an UltraScale FPGA," *IEEE Trans. Nucl. Sci.*, vol. 63, no. 5, pp. 2617–2621, Oct. 2016, doi: 10.1109/TNS.2016.2596305.

[13] D. Chaberski, "Time-to-digital-converter based on multiple-tapped-delay-line," *Measurement*, vol. 89, pp. 87–96, Jul. 2016, doi: 10.1016/j.measurement.2016.03.065.

[14] Y. Wang, W. Xie, H. Chen, and D. Day-Uei Li, "High-resolution time-to-digital converters (TDCs) with a bidirectional encoder," *Measurement*, vol. 206, Jan. 2023, Art. no. 112258, doi: 10.1016/j.measurement.2022.112258.

[15] M. P. Mattada and H. Guhilot, "Time-to-digital converters—A comprehensive review," *Int. J. Circuit Theory Appl.*, vol. 49, no. 3, pp. 778–800, Mar. 2021, doi: 10.1002/cta.2936.

[16] R. Scott, W. Jiang, and M. J. Deen, "CMOS time-to-digital converters for biomedical imaging applications," *IEEE Rev. Biomed. Eng.*, vol. 16, pp. 627–652, 2023, doi: 10.1109/RBME.2021.3092197.

[17] F. Piron, D. Morrison, M. R. Yuce, and J.-M. Redouté, "A review of single-photon avalanche diode time-of-flight imaging sensor arrays," *IEEE Sensors J.*, vol. 21, no. 11, pp. 12654–12666, Jun. 2021, doi: 10.1109/JSEN.2020.3039362.

[18] R. Machado, J. Cabral, and F. S. Alves, "Recent developments and challenges in FPGA-based time-to-digital converters," *IEEE Trans. Instrum. Meas.*, vol. 68, no. 11, pp. 4205–4221, Nov. 2019, doi: 10.1109/TIM.2019.2938436.

[19] A. Tontini, L. Gasparini, L. Pancheri, and R. Passerone, "Design and characterization of a low-cost FPGA-based TDC," *IEEE Trans. Nucl. Sci.*, vol. 65, no. 2, pp. 680–690, Feb. 2018, doi: 10.1109/TNS.2018.2790703.

[20] F. Garzetti, N. Corna, N. Lusardi, and A. Geraci, "Time-to-digital converter IP-core for FPGA at state of the art," *IEEE Access*, vol. 9, pp. 85515–85528, 2021, doi: 10.1109/ACCESS.2021.3088448.

[21] X. Hu, L. Zhao, S. Liu, J. Wang, and Q. An, "A stepped-up tree encoder for the 10-ps wave union TDC," in *Proc. 18th IEEE-NPSS Real Time Conf.*, Jun. 2012, pp. 1–5, doi: 10.1109/RTC.2012.6418105.

[22] Y. Wang, C. Liu, X. Cheng, and D. Li, "Spartan-6 FPGA based 8-channel time-to-digital converters for TOF-PET systems," in *Proc. IEEE Nucl. Sci. Symp. Med. Imag. Conf. (NSS/MIC)*, Oct. 2015, pp. 1–3, doi: 10.1109/NSSMIC.2015.7582016.

[23] W. Xie, H. Chen, and D. D. Li, "Efficient time-to-digital converters in 20 nm FPGAs with wave union methods," *IEEE Trans. Ind. Electron.*, vol. 69, no. 1, pp. 1021–1031, Jan. 2022, doi: 10.1109/TIE.2021.3053905.

[24] Y. Wang, J. Kuang, C. Liu, Q. Cao, and D. Li, "A flexible 32-channel time-to-digital converter implemented in a Xilinx Zynq-7000 field programmable gate array," *Nucl. Instrum. Methods Phys. Res. A, Accel. Spectrom. Detect. Assoc. Equip.*, vol. 847, pp. 61–66, Mar. 2017, doi: 10.1016/j.nima.2016.11.031.

[25] Q. Cao, Y. Wang, and C. Liu, "A combination of multiple channels of FPGA based time-to-digital converter for high time precision," in *Proc. IEEE Nucl. Sci. Symp., Med. Imag. Conf. Room-Temperature Semiconductor Detect. Workshop (NSS/MIC/RTSD)*, Oct. 2016, pp. 1–3, doi: 10.1109/NSSMIC.2016.8069649.

[26] W. Pan, G. Gong, H. Li, and J. Li, "A 20-ps temperature compensated time-to-digital converter (TDC) implemented in FPGA," in *Proc. IEEE Nucl. Sci. Symp. Med. Imag. Conf. (NSS/MIC)*, Oct. 2013, pp. 1–6, doi: 10.1109/NSSMIC.2013.6829535.

[27] N. Lusardi, A. Abba, F. Caponio, and A. Geraci, "Quantization noise in non-homogeneous calibration table of a TCD implemented in FPGA," in *Proc. IEEE Nucl. Sci. Symp. Med. Imag. Conf. (NSS/MIC)*, Nov. 2014, pp. 1–5, doi: 10.1109/NSSMIC.2014.7431149.

[28] N. Lusardi, F. Garzetti, and A. Geraci, "Digital instrument with configurable hardware and firmware for multi-channel time measures," *Rev. Sci. Instrum.*, vol. 90, no. 5, May 2019, Art. no. 055113, doi: 10.1063/1.5028131.

[29] P. Kwiatkowski and R. Szplet, "Multisampling wave union time-to-digital converter," in *Proc. 6th Int. Conf. Event-Based Control, Commun., Signal Process. (EBCCSP)*, Sep. 2020, pp. 1–5, doi: 10.1109/EBCCSP51266.2020.9291363.

[30] X. Deng and Q. Chen, "A 4.32-ps precision time-to-digital convertor using multisampling wave union method on a 28-nm FPGA," *J. Instrum.*, vol. 16, no. 12, Dec. 2021, Art. no. P12031, doi: 10.1088/1748-0221/16/12/p12031.

[31] P. Carra et al., "Auto-calibrating TDC for an SoC-FPGA data acquisition system," *IEEE Trans. Radiat. Plasma Med. Sci.*, vol. 3, no. 5, pp. 549–556, Sep. 2019, doi: 10.1109/TRPMS.2018.2882709.

[32] W. Zhang et al., "A low-power time-to-digital converter for the CMS endcap timing layer (ETL) upgrade," *IEEE Trans. Nucl. Sci.*, vol. 68, no. 8, pp. 1984–1992, Aug. 2021, doi: 10.1109/TNS.2021.3085564.

[33] X. Yu, S. Chang, W. Li, and H. Xia, "A low-cost FPGA-based coarse-fine counting time-to-digital converter with external high-precision reference clock," *IEEE Trans. Instrum. Meas.*, vol. 72, pp. 1–10, 2023, doi: 10.1109/TIM.2023.3277958.

[34] M. D. Fries and J. J. Williams, "High-precision TDC in an FPGA using a 192 MHz quadrature clock," in *Proc. IEEE Nucl. Sci. Symp. Conf. Rec.*, Nov. 2002, pp. 580–584, doi: 10.1109/NSSMIC.2002.1239380.

[35] R. Szplet, J. Kalisz, and R. Szymanowski, "Interpolating time counter with 100 ps resolution on a single FPGA device," *IEEE Trans. Instrum. Meas.*, vol. 49, no. 4, pp. 879–883, Apr. 2000, doi: 10.1109/19.863942.

[36] A. Aloisio et al., "High-resolution time-to-digital converter in field programmable gate array," *Genetic Algorithms Field Program.*, vol. 8, pp. 383–387, 2008, doi: 10.5170/CERN-2008-008.383.

[37] N. Franch, O. Alonso, J. Canals, A. Vilá, and A. Dieguez, "A low cost fluorescence lifetime measurement system based on SPAD detectors and FPGA processing," *J. Instrum.*, vol. 12, no. 2, Feb. 2017, Art. no. C02070, doi: 10.1088/1748-0221/12/02/c02070.

[38] A. O. Korkan and H. Yuksel, "A novel time-to-amplitude converter and a low-cost wide dynamic range FPGA TDC for LiDAR application," *IEEE Trans. Instrum. Meas.*, vol. 71, pp. 1–15, 2022, doi: 10.1109/TIM.2022.3200117.

[39] S. S. Junnarkar, M. Purschke, J.-F. Pratte, S.-J. Park, P. O'Connor, and R. Fontaine, "An FPGA-based, 12-channel TDC and digital signal processing module for the RatCAP scanner," in *Proc. IEEE Nucl. Sci. Symp. Conf. Rec.*, Nov. 2005, pp. 919–923, doi: 10.1109/NSSMIC.2005.1596404.

[40] C. Kai et al., "A high precision time-to-digital converter based on multi-phase clock implemented within field-programmable-gate-array," *Nucl. Sci. Techn.*, vol. 25, no. 11, pp. 1014–1025, 2010, doi: 10.1142/S0217732310000174.

[41] T. Suwada, F. Miyahara, K. Furukawa, M. Shoji, M. Ikeno, and M. Tanaka, "Wide dynamic range FPGA-based TDC for monitoring a trigger timing distribution system in linear accelerators," *Nucl. Instrum. Methods Phys. Res. A, Accel. Spectrom. Detect. Assoc. Equip.*, vol. 786, pp. 83–90, Jun. 2015, doi: 10.1016/j.nima.2015.03.019.

[42] S. M. Patanwala et al., "A high-throughput photon processing technique for range extension of SPAD-based LiDAR receivers," *IEEE Open J. Solid-State Circuits Soc.*, vol. 2, pp. 12–25, 2022, doi: 10.1109/OJSSCS.2021.3118987.

[43] D. F. Spencer, J. Cole, M. Drigert, and R. Aryaeinejad, "A high-resolution, multi-stop, time-to-digital converter for nuclear time-of-flight measurements," *Nucl. Instrum. Methods Phys. Res. A, Accel. Spectrom. Detect. Assoc. Equip.*, vol. 556, no. 1, pp. 291–295, Jan. 2006, doi: 10.1016/j.nima.2005.10.008.

[44] Y. Wang, P. Kuang, and C. Liu, "A 256-channel multi-phase clock sampling-based time-to-digital converter implemented in a Kintex-7 FPGA," in *Proc. IEEE Int. Instrum. Meas. Technol. Conf.*, May 2016, pp. 1–5, doi: 10.1109/I2MTC.2016.7520401.

[45] Z. Li et al., "Development of an integrated four-channel fast avalanche-photodiode detector system with nanosecond time resolution," *Nucl. Instrum. Methods Phys. Res. A, Accel. Spectrom. Detect. Assoc. Equip.*, vol. 870, pp. 43–49, Oct. 2017, doi: 10.1016/j.nima.2017.05.021.

[46] F. S. Deng, H. Liang, J. Y. Tang, and B. J. Ye, "Design of a 32-channel NIM TDC module based on single FPGA for the μSR spectrometer prototype at the China spallation neutron source," *J. Instrum.*, vol. 14, no. 6, Jun. 2019, Art. no. T06002, doi: 10.1088/1748-0221/14/06/t06002.

[47] J. Wu, S. Hansen, and Z. Shi, "ADC and TDC implemented using FPGA," in *Proc. IEEE Nucl. Sci. Symp. Conf. Rec.*, Oct. 2007, pp. 281–286, doi: 10.1109/NSSMIC.2007.4436331.

[48] M. Büchele, H. Fischer, F. Herrmann, K. Königsmann, C. Schill, and S. Schopferer, "The GANDALF 128-channel time-to-digital converter," *Phys. Proc.*, vol. 37, no. 3, pp. 1827–1834, 2012, doi: 10.1016/j.phpro.2012.02.504.

[49] Z. Yin, S. Liu, X. Hao, S. Gao, and Q. An, "A high-resolution time-to-digital converter based on multi-phase clock implement in field-programmable-gate-array," in *Proc. 18th IEEE-NPSS Real Time Conf.*, Jun. 2012, pp. 1–4, doi: 10.1109/RTC.2012.6418217.

[50] L. Iafolla et al., "FPGA-based time to digital converter and data acquisition system for high energy tagger of KLOE-2 experiment," *Nucl. Instrum. Methods Phys. Res. A, Accel. Spectrom. Detect. Assoc. Equip.*, vol. 718, pp. 213–214, Aug. 2013, doi: 10.1016/j.nima.2012.10.010.

[51] T. Xiang et al., "A 56-ps multi-phase clock time-to-digital convertor based on Artix-7 FPGA," in *Proc. 19th IEEE-NPSS Real Time Conf.*, May 2014, pp. 1–4, doi: 10.1109/RTC.2014.7097544.

[52] A. Balla et al., "The characterization and application of a low resource FPGA-based time to digital converter," *Nucl. Instrum. Methods Phys. Res. A, Accel. Spectrom. Detect. Assoc. Equip.*, vol. 739, pp. 75–82, Mar. 2014, doi: 10.1016/j.nima.2013.12.033.

[53] D. Calvo, "1 ns time to digital converters for the KM3NeT data readout system," *AIP Conf. Proc.*, vol. 1631, pp. 98–101, Nov. 2014, doi: 10.1063/1.4902781.

[54] Y. Sano, Y. Horii, M. Ikeno, O. Sasaki, M. Tomoto, and T. Uchida, "Subnanosecond time-to-digital converter implemented in a Kintex-7 FPGA," *Nucl. Instrum. Methods Phys. Res. A, Accel. Spectrom. Detect. Assoc. Equip.*, vol. 874, pp. 50–56, Dec. 2017, doi: 10.1016/j.nima.2017.08.038.

[55] P. Kwiatkowski, K. Róüyc, M. Sawicki, Z. Jachna, and R. Szplet, "5 ps jitter programmable time Interval/Frequency generator," *Metrol. Meas. Syst.*, vol. 24, no. 1, pp. 57–68, Mar. 2017, doi: 10.1515/mms-2017-0009.

[56] R. Szplet, J. Kalisz, and Z. Jachna, "A 45 ps time digitizer with a two-phase clock and dual-edge two-stage interpolation in a field programmable gate array device," *Meas. Sci. Technol.*, vol. 20, no. 2, Feb. 2009, Art. no. 025108, doi: 10.1088/0957-0233/20/2/025108.

[57] M. Zieliński, D. Chaberski, M. Kowalski, R. Frankowski, and S. Grzelak, "High-resolution time-interval measuring system implemented in single FPGA device," *Measurement*, vol. 35, no. 3, pp. 311–317, Apr. 2004, doi: 10.1016/j.measurement.2003.12.001.

[58] Y. Wang, J. Kuang, C. Liu, and Q. Cao, "A 3.9-ps RMS precision time-to-digital converter using ones-counter encoding scheme in a Kintex-7 FPGA," *IEEE Trans. Nucl. Sci.*, vol. 64, no. 10, pp. 2713–2718, Oct. 2017, doi: 10.1109/TNS.2017.2746626.

[59] Y. Wang, Q. Cao, and C. Liu, "A multi-chain merged tapped delay line for high precision time-to-digital converters in FPGAs," *IEEE Trans. Circuits Syst. II, Exp. Briefs*, vol. 65, no. 1, pp. 96–100, Jan. 2018, doi: 10.1109/TCSII.2017.2698479.

[60] J. Wang, C. Feng, W. Dong, Z. Shen, and S. Liu, "A high precision time-to-digital converter based on multi-chain interpolation with a low cost Artix-7 FPGA," in *Proc. 7th Int. Conf. Event-Based Control, Commun., Signal Process. (EBCCSP)*, Jun. 2021, pp. 1–5, doi: 10.1109/EBCCSP53293.2021.9502368.

[61] J. Kalisz, R. Szplet, J. Pasierbinski, and A. Poniecki, "Field-programmable-gate-array-based time-to-digital converter with 200-ps resolution," *IEEE Trans. Instrum. Meas.*, vol. 46, no. 1, pp. 51–55, Mar. 1997, doi: 10.1109/19.552156.

[62] R. Jiang et al., "Successive approximation time-to-digital converter with Vernier-level resolution," in *IEEE MTT-S Int. Microw. Symp. Dig.*, Jul. 2016, pp. 1–6, doi: 10.1109/IMS3TW.2016.7524226.

[63] L. Bengtsson, "Embedded Vernier TDC with sub-nano second resolution using fractional-*N* PLL," *Measurement*, vol. 108, pp. 48–54, Oct. 2017, doi: 10.1016/j.measurement.2017.05.038.

[64] Y. Jiao, X. Shi, L. Zhou, W. Chen, and C. Chen, "A Vernier caliper time-to-digital converters with ultralow nonlinearity in FPGAs," in *Proc. IEEE 20th Int. Conf. Commun. Technol. (ICCT)*, Oct. 2020, pp. 1655–1659, doi: 10.1109/ICCT50939.2020.9295834.

[65] A. M. Amiri, A. Khouas, and M. Boukadoum, "On the timing uncertainty in delay-line-based time measurement applications targeting FPGAs," in *Proc. IEEE Int. Symp. Circuits Syst. (ISCAS)*, May 2007, pp. 3772–3775, doi: 10.1109/ISCAS.2007.378782.

[66] M.-C. Lin, G.-R. Tsai, C.-Y. Liu, and S.-S. Chu, "FPGA-based high area efficient time-to-digital IP design," in *Proc. IEEE Region 10 Conf.*, Nov. 2006, pp. 1–4, doi: 10.1109/TENCON.2006.343706.

[67] K. Katoh et al., "A small chip area stochastic calibration for TDC using ring oscillator," *J. Electron. Test.*, vol. 30, no. 6, pp. 653–663, Dec. 2014, doi: 10.1007/s10836-014-5486-0.

[68] M. Abbas and K. Khalil, "A 23 ps resolution time-to-digital converter implemented on low-cost FPGA platform," in *Proc. Int. Symp. Signals, Circuits Syst. (ISSCS)*, Jul. 2015, pp. 1–4, doi: 10.1109/ISSCS.2015.7203949.

[69] S. Berrima, Y. Blaquière, and Y. Savaria, "A multi-measurements RO-TDC implemented in a Xilinx field programmable gate array," in *Proc. IEEE Int. Symp. Circuits Syst. (ISCAS)*, May 2017, pp. 1–4, doi: 10.1109/ISCAS.2017.8050436.

[70] K. Cui, Z. Ren, X. Li, Z. Liu, and R. Zhu, "A high-linearity, ring-oscillator-based, Vernier time-to-digital converter utilizing carry chains in FPGAs," *IEEE Trans. Nucl. Sci.*, vol. 64, no. 1, pp. 697–704, Jan. 2017, doi: 10.1109/TNS.2016.2632168.

[71] J. Kuang, Y. Wang, Q. Cao, and C. Liu, "Implementation of a high precision multi-measurement time-to-digital convertor on a Kintex-7 FPGA," *Nucl. Instrum. Methods Phys. Res. A, Accel. Spectrom. Detect. Assoc. Equip.*, vol. 891, pp. 37–41, May 2018, doi: 10.1016/j.nima.2018.02.064.

[72] V. L. Dinh, X. T. Nguyen, and H.-J. Lee, "A novel FPGA implementation of a time-to-digital converter supporting run-time estimation and compensation," *ACM Trans. Reconfigurable Technol. Syst.*, vol. 12, no. 2, pp. 1–21, Jun. 2019, doi: 10.1145/3322482.

[73] A. M. Z. Khaki, E. Farshidi, and K. A. Asl, "Design and FPGA implementation of a multirate $\Delta\Sigma$ time-to-digital converter with third-order noise-shaping," *Microelectron. J.*, vol. 108, Feb. 2021, Art. no. 104982, doi: 10.1016/j.mejo.2020.104982.

[74] D. Li, J. Hu, R. Ma, X. Wang, Y. Liu, and Z. Zhu, "SPAD-based LiDAR with real-time accuracy calibration and laser power regulation," *IEEE Trans. Circuits Syst. II, Exp. Briefs*, vol. 70, no. 2, pp. 431–435, Feb. 2023, doi: 10.1109/TCSII.2022.3211868.

[75] S. Berrima, Y. Blaquière, and Y. Savaria, "Ring-oscillator-based high accuracy low complexity multichannel time-to-digital converter architecture for field-programmable gate arrays," *IEEE Trans. Instrum. Meas.*, vol. 70, pp. 1–10, 2021, doi: 10.1109/TIM.2021.3106100.

[76] J. Zhang, Y. Wang, and Z. Song, "A ring-oscillator based multi-mode time-to-digital converter on Xilinx Kintex-7 FPGA," *Nucl. Instrum. Methods Phys. Res. A, Accel. Spectrom. Detect. Assoc. Equip.*, vol. 1011, Sep. 2021, Art. no. 165578, doi: 10.1016/j.nima.2021.165578.

[77] M. S. Andaloussi, M. Boukadoum, and E. M. Aboulhamid, "A novel time-to-digital converter with 150 ps time resolution and 2.5 ns pulse-pair resolution," in *Proc. 14th Int. Conf. Microelectron.*, Dec. 2003, pp. 123–126, doi: 10.1109/ICM-02.2002.1161511.

[78] A. Amiri, M. Boukadoum, and A. Khouas, "Low dead time, multi-hit fpga-based time-to-digital converter," in *Proc. IEEE North-East Workshop Circuits Syst.*, Jun. 2006, pp. 29–32, doi: 10.1109/NEWCAS.2006.250905.

[79] P. Chen et al., "High-precision PLL delay matrix with overclocking and double data rate for accurate FPGA time-to-digital converters," *IEEE Trans. Very Large Scale Integr. (VLSI) Syst.*, vol. 28, no. 4, pp. 904–913, Apr. 2020, doi: 10.1109/TVLSI.2019.2962606.

[80] N. Lusardi et al., "High-resolution imager based on time-to-space conversion," *IEEE Trans. Instrum. Meas.*, vol. 71, pp. 1–11, 2022, Art. no. 2004811, doi: 10.1109/TIM.2022.3198442.

[81] A. Aloisio, P. Branchini, R. Giordano, V. Izzo, and S. Loffredo, "High-precision time-to-digital converter in a FPGA device," in *Proc. 16th IEEE-NPSS Real Time Conf.*, May 2009, pp. 283–286, doi: 10.1109/RTC.2009.5322005.

[82] H. Chen and D. D. Li, "Multichannel, low nonlinearity time-to-digital converters based on 20 and 28 nm FPGAs," *IEEE Trans. Ind. Electron.*, vol. 66, no. 4, pp. 3265–3274, Apr. 2019, doi: 10.1109/TIE.2018.2842787.

[83] L. Shubin, W. Jinong, C. Kai, and A. Qi, "Double sampling architecture for performing fine time interpolation within a limited logic density FPGA," in *Proc. 9th Int. Conf. Electron. Meas. Instrum.*, Aug. 2009, pp. 1-585–1-589, doi: 10.1109/icemi.2009.5274799.

[84] Y.-H. Chen, "A high resolution FPGA-based merged delay line TDC with nonlinearity calibration," in *Proc. IEEE Int. Symp. Circuits Syst. (ISCAS)*, May 2013, pp. 2432–2435, doi: 10.1109/ISCAS.2013.6572370.

[85] Y.-H. Chen, "A counting-weighted calibration method for a field-programmable-gate-array-based time-to-digital converter," *Nucl. Instrum. Methods Phys. Res. A, Accel. Spectrom. Detect. Assoc. Equip.*, vol. 854, pp. 61–63, May 2017, doi: 10.1016/j.nima.2017.02.053.

[86] P. Kwiatkowski and R. Szplet, "Time-to-digital converter with pseudo-segmented delay line," in *Proc. IEEE Int. Instrum. Meas. Technol. Conf. (I2MTC)*, May 2019, pp. 1–6, doi: 10.1109/I2MTC.2019.8826927.

[87] P. Kwiatkowski and R. Szplet, "Efficient implementation of multiple time coding lines-based TDC in an FPGA device," *IEEE Trans. Instrum. Meas.*, vol. 69, no. 10, pp. 7353–7364, Oct. 2020, doi: 10.1109/TIM.2020.2984929.

[88] J. Deng, P. Yin, X. Lei, Z. Shu, M. Tang, and F. Tang, "A tunable parameter, high linearity time-to-digital converter implemented in 28-nm FPGA," *IEEE Trans. Instrum. Meas.*, vol. 70, pp. 1–12, 2021, doi: 10.1109/TIM.2021.3117373.

[89] Y. Hua and D. Chitnis, "A highly linear and flexible FPGA-based time-to-digital converter," *IEEE Trans. Ind. Electron.*, vol. 69, no. 12, pp. 13744–13753, Dec. 2022, doi: 10.1109/TIE.2021.3128912.

[90] Y. Wang, W. Xie, H. Chen, and D. D.-U. Li, "Low-hardware consumption, resolution-configurable gray code oscillator time-to-digital converters implemented in 16 nm, 20 nm, and 28 nm FPGAs," *IEEE Trans. Ind. Electron.*, vol. 70, no. 4, pp. 4256–4266, Apr. 2023, doi: 10.1109/TIE.2022.3174299.

[91] M. Parsakordasiabi, I. Vornicu, Á. Rodríguez-Vázquez, and R. Carmona-Galán, "An efficient TDC using a dual-mode resource-saving method evaluated in a 28-nm FPGA," *IEEE Trans. Instrum. Meas.*, vol. 71, pp. 1–13, 2022, doi: 10.1109/TIM.2021.3136267.

[92] J. Wu, Z. Shi, and I. Y. Wang, "Firmware-only implementation of time-to-digital converter (TDC) in field-programmable gate array (FPGA)," in *Proc. IEEE Nucl. Sci. Symposium. Conf. Rec.*, Oct. 2003, pp. 177–181, doi: 10.1109/NSSMIC.2003.1352025.

[93] C. Favi and E. Charbon, "A 17 ps time-to-digital converter implemented in 65nm FPGA technology," in *Proc. ACM/SIGDA Int. Symp. Field Program. Gate Arrays*, Feb. 2009, pp. 113–120, doi: 10.1145/1508128.1508145.

[94] L. Zhao et al., "The design of a 16-channel 15 ps TDC implemented in a 65 nm FPGA," *IEEE Trans. Nucl. Sci.*, vol. 60, no. 5, pp. 3532–3536, Oct. 2013, doi: 10.1109/TNS.2013.2280909.

[95] X. Kong, Y. Wang, Z. Song, X. Zhou, J. Lin, and J. Kuang, "A resource-saving method for implementation of high-performance time-to-digital converters in FPGA," in *Proc. IEEE Int. Instrum. Meas. Technol. Conf. (I2MTC)*, May 2020, pp. 1–5, doi: 10.1109/I2MTC43012.2020.9129275.

[96] P. Kwiatkowski, D. Sondej, and R. Szplet, "Subpicosecond resolution time interval counter with multisampling wave union type b TDCs in 28 nm FPGA device," *Measurement*, vol. 209, Mar. 2023, Art. no. 112510, doi: 10.1016/j.measurement.2023.112510.

[97] N. Dutton et al., "Multiple-event direct to histogram TDC in 65 nm FPGA technology," in *Proc. IEEE Ph.D Res. Microelectron. Electron.*, Jun. 2014, pp. 1–5, doi: 10.1109/PRIME.2014.6872727.

[98] Y. Bai et al., "Intelligent FPGA data acquisition framework," *IEEE Trans. Nucl. Sci.*, vol. 64, no. 6, pp. 1219–1224, Jun. 2017, doi: 10.1109/TNS.2017.2708510.

[99] Y. Jiao, Q. Zhang, W. Chen, L. Zhou, and C. Chen, "A resource-saving dual channel time-to-digital converter with shared tapped delay line in FPGAs," *J. Instrum.*, vol. 16, no. 1, Jan. 2021, Art. no. P01012, doi: 10.1088/1748-0221/16/01/p01012.

[100] J. Y. Won, S. I. Kwon, H. S. Yoon, G. B. Ko, J.-W. Son, and J. S. Lee, "Dual-phase tapped-delay-line time-to-digital converter with on-the-fly calibration implemented in 40 nm FPGA," *IEEE Trans. Biomed. Circuits Syst.*, vol. 10, no. 1, pp. 231–242, Feb. 2016, doi: 10.1109/TBCAS.2015.2389227.

[101] H. Chen, Y. Zhang, and D. D.-U. Li, "A low nonlinearity, missing-code free time-to-digital converter based on 28-nm FPGAs with embedded bin-width calibrations," *IEEE Trans. Instrum. Meas.*, vol. 66, no. 7, pp. 1912–1921, Jul. 2017, doi: 10.1109/TIM.2017.2663498.

[102] J. Wu, "Several key issues on implementing delay line based TDCs using FPGAs," *IEEE Trans. Nucl. Sci.*, vol. 57, no. 3, pp. 1543–1548, Jun. 2010, doi: 10.1109/TNS.2010.2045901.

[103] J. Zheng, P. Cao, D. Jiang, and Q. An, "Low-cost FPGA TDC with high resolution and density," *IEEE Trans. Nucl. Sci.*, vol. 64, no. 6, pp. 1401–1408, Jun. 2017, doi: 10.1109/TNS.2017.2705802.

[104] G. Cao, H. Xia, and N. Dong, "An 18-ps TDC using timing adjustment and bin realignment methods in a cyclone-IV FPGA," *Rev. Sci. Instrum.*, vol. 89, no. 5, May 2018, Art. no. 054707, doi: 10.1063/1.5008610.

[105] H. Xia, G. Cao, and N. Dong, "A 6.6 ps RMS resolution time-to-digital converter using interleaved sampling method in a 28 nm FPGA," *Rev. Sci. Instrum.*, vol. 90, no. 4, Apr. 2019, Art. no. 044706, doi: 10.1063/1.5084014.

[106] W. Xie, Y. Wang, H. Chen, and D. D. Li, "128-channel high-linearity resolution-adjustable time-to-digital converters for LiDAR applications: Software predictions and hardware implementations," *IEEE Trans. Ind. Electron.*, vol. 69, no. 4, pp. 4264–4274, Apr. 2022, doi: 10.1109/TIE.2021.3076708.

[107] Y. Wang and C. Liu, "A nonlinearity minimization-oriented resource-saving time-to-digital converter implemented in a 28 nm Xilinx FPGA," *IEEE Trans. Nucl. Sci.*, vol. 62, no. 5, pp. 2003–2009, Oct. 2015, doi: 10.1109/TNS.2015.2475630.

[108] R. Szymanowski and J. Kalisz, "Field programmable gate array time counter with two-stage interpolation," *Rev. Scientific Instrum.*, vol. 76, no. 4, Apr. 2005, Art. no. 045104, doi: 10.1063/1.1878212.

[109] J. Wang, S. Liu, Q. Shen, H. Li, and Q. An, "A fully fledged TDC implemented in field-programmable gate arrays," *IEEE Trans. Nucl. Sci.*, vol. 57, no. 2, pp. 446–450, Apr. 2010, doi: 10.1109/TNS.2009.2037958.

[110] S. Rogacki and T. H. Zurbuchen, "A time digitizer for space instrumentation using a field programmable gate array," *Rev. Sci. Instrum.*, vol. 84, no. 8, Aug. 2013, Art. no. 083107, doi: 10.1063/1.4818965.

[111] S. Zhao, Z. Wang, M. Li, P. Zhang, and J. Cheng, "FPGA-only MVT digitizer for neutron downhole applications," *Measurement*, vol. 211, Apr. 2023, Art. no. 112649, doi: 10.1016/j.measurement.2023.112649.

[112] K.-J. Choi and D.-W. Jee, "Design and calibration techniques for a multichannel FPGA-based time-to-digital converter in an object positioning system," *IEEE Trans. Instrum. Meas.*, vol. 70, pp. 1–9, 2021, doi: 10.1109/TIM.2020.3011490.

[113] Y. Wang, W. Xie, H. Chen, and D. D. Li, "Multichannel time-to-digital converters with automatic calibration in Xilinx Zynq-7000 FPGA devices," *IEEE Trans. Ind. Electron.*, vol. 69, no. 9, pp. 9634–9643, Sep. 2022, doi: 10.1109/TIE.2021.3111563.

[114] H. Kleines, A. Ackens, P. Kämmerling, M. Drochner, P. Wüstner, and W. Erven, "Development of a high resolution TDC module for MicroTCA based on the GPX ASIC," in *Proc. IEEE Nucl. Sci. Symposuim Med. Imag. Conf.*, Oct. 2010, pp. 755–757, doi: 10.1109/NSSMIC.2010.5873859.

[115] Y. Huang, S. Liu, J. Wang, X. Hu, C. Feng, and Q. An, "Development of a high resolution PXI based data acquisition system for electron momentum spectrometer," in *Proc. 18th IEEE-NPSS Real Time Conf.*, Jun. 2012, pp. 1–4, doi: 10.1109/RTC.2012.6418154.

[116] R. Narasimman, A. Prabhakar, and N. Chandrachoodan, "Implementation of a 30 ps resolution time to digital converter in FPGA," in *Proc. Int. Conf. Electron. Design, Comput. Netw. Automated Verification (EDCAV)*, Jan. 2015, pp. 12–17, doi: 10.1109/EDCAV.2015.7060530.

[117] T. Sui, Z. Zhao, S. Xie, Q. Huang, J. Xu, and Q. Peng, "Pico-TDC: A novel FPGA-based TDC with 2.2 ps RMS timing resolution," in *Proc. IEEE Nucl. Sci. Symp. Med. Imag. Conf. (NSS/MIC)*, Oct. 2017, pp. 1–3, doi: 10.1109/NSSMIC.2017.8532955.

[118] M. Adamic and A. Trost, "A fast high-resolution time-to-digital converter implemented in a Zynq 7010 SoC," in *Proc. Austrochip Workshop Microelectron. (Austrochip)*, Oct. 2019, pp. 29–34, doi: 10.1109/Austrochip.2019.00017.

[119] S. M. Ramzy and K. Hares, "High resolution time-to-digital converter using low resources FPGA for time-of-flight measurement," *Microelectron. J.*, vol. 101, Jul. 2020, Art. no. 104822, doi: 10.1016/j.mejo.2020.104822.

[120] V. Bocci, G. Chiodi, F. Iacoangeli, R. Nobrega, D. Pinci, and W. Rinaldi, "Time-multiplexing of signals using highly integrated digital delay: An FPGA implementation," in *Proc. IEEE Nucl. Sci. Symp. Conf. Rec.*, Oct. 2005, pp. 398–402, doi: 10.1109/NSSMIC.2005.1596279.

[121] V. Bocci, F. Iacoangeli, and R. Nobrega, "A multichannel TDC based on ring-delay time multiplexer: A prototype," in *Proc. IEEE Nucl. Sci. Symp. Conf. Rec.*, Oct. 2007, pp. 720–724, doi: 10.1109/NSSMIC.2007.4436433.

[122] D. Kim, Y. Choi, and S. Lee, "An improved method of FPGA-based TDC for time-of-flight PET," in *Proc. IEEE Nucl. Sci. Symp. Med. Imag. Conf. (NSS/MIC)*, Oct. 2013, pp. 1–3, doi: 10.1109/NSSMIC.2013.6829201.

[123] Z. Song, Y. Wang, and J. Kuang, "Implementation of 5.3 ps RMS precision and 350 m samples/second throughput time-to-digital converters with event sampling architecture in a Kintex-7 FPGA," *Nucl. Instrum. Methods Phys. Res. A, Accel. Spectrom. Detect. Assoc. Equip.*, vol. 944, Nov. 2019, Art. no. 162584, doi: 10.1016/j.nima.2019.162584.

[124] M. W. Fishburn, L. H. Menninga, C. Favi, and E. Charbon, "A 19.6 ps, FPGA-based TDC with multiple channels for open source applications," *IEEE Trans. Nucl. Sci.*, vol. 60, no. 3, pp. 2203–2208, Jun. 2013, doi: 10.1109/TNS.2013.2241789.

[125] L.-Y. Hsu and J.-L. Huang, "A multi-channel FPGA-based time-to-digital converter," in *IEEE MTT-S Int. Microw. Symp. Dig.*, Jul. 2016, pp. 1–4, doi: 10.1109/IMS3TW.2016.7524227.

[126] S. Burri et al., "LinoSPAD: A time-resolved 256 × 1 CMOS SPAD line sensor system featuring 64 FPGA-based TDC channels running at up to 8.5 giga-events per second," *Proc. SPIE*, vol. 9899, Apr. 2016, Art. no. 98990D, doi: 10.1117/12.2227564.

[127] X. Yu, H. Xia, W. Li, J. Zhang, and S. Chang, "A 4.8 ps root-mean-square resolution time-to-digital converter implemented in a 20 nm cyclone-10 GX field-programmable gate array," *Rev. Sci. Instrum.*, vol. 93, no. 8, Aug. 2022, Art. no. 085001, doi: 10.1063/5.0090783.

[128] J. Kuang, Y. Wang, and C. Liu, "A 128-channel high performance time-to-digital converter implemented in an UltraScale FPGA," in *Proc. IEEE Nucl. Sci. Symp. Med. Imag. Conf. (NSS/MIC)*, Oct. 2017, pp. 1–4, doi: 10.1109/NSSMIC.2017.8532676.

[129] N. Lusardi, F. Garzetti, N. Corna, R. D. Marco, and A. Geraci, "Very high-performance 24-channels time-to-digital converter in Xilinx 20-nm kintex UltraScale FPGA," in *Proc. IEEE Nucl. Sci. Symp. Med. Imag. Conf. (NSS/MIC)*, Oct. 2019, pp. 1–4, doi: 10.1109/NSS/MIC42101.2019.9059958.

[130] H.-B. Xie, Y. Li, Q. Shen, S.-K. Liao, and C.-Z. Peng, "A high-precision 2.5-ps RMS time synchronization for multiple high-speed transceivers in FPGA," *IEEE Trans. Nucl. Sci.*, vol. 66, no. 7, pp. 1070–1075, Jul. 2019, doi: 10.1109/TNS.2019.2904703.

[131] Y.-H. Chen, "Dual-mode FPGA-based triple-TDC with real-time calibration and a triple modular redundancy scheme," *Electronics*, vol. 9, no. 4, p. 607, Apr. 2020, doi: 10.3390/electronics9040607.

[132] R. Machado, L. A. Rocha, and J. Cabral, "A novel synchronizer for a 17.9 ps Nutt time-to-digital converter implemented on FPGA," in *Proc. AEIT Int. Annu. Conf.*, Oct. 2018, pp. 1–6, doi: 10.23919/AEIT.2018.8577365.

[133] M. Zhang, H. Wang, and B. Liu, "A high precision TDC design based on FPGA+ARM," *J. Phys., Conf. Ser.*, vol. 1486, no. 7, Apr. 2020, Art. no. 072054, doi: 10.1088/1742-6596/1486/7/072054.

[134] Z. Song et al., "An 8.8 ps RMS resolution time-to-digital converter implemented in a 60 nm FPGA with real-time temperature correction," *Sensors*, vol. 20, no. 8, p. 2172, Apr. 2020, doi: 10.3390/s20082172.

[135] W. Khaddour, F. Dadouche, W. Uhring, V. Frick, and M. Madec, "Design methodology and timing considerations for implementing a TDC on a cyclone V FPGA target," in *Proc. 18th IEEE Int. New Circuits Syst. Conf. (NEWCAS)*, Jun. 2020, pp. 126–129, doi: 10.1109/NEWCAS49341.2020.9159812.

[136] W. Xie, H. Chen, Z. Zang, and D. D. Li, "Multi-channel high-linearity time-to-digital converters in 20 nm and 28 nm FPGAs for LiDAR applications," in *Proc. 6th Int. Conf. Event-Based Control, Commun., Signal Process. (EBCCSP)*, Sep. 2020, pp. 1–4, doi: 10.1109/EBCCSP51266.2020.9291360.

[137] Y. Tang, T. Townsend, H. Deng, Y. Liu, R. Zhang, and J. Chen, "A highly linear FPGA-based TDC and a low-power multichannel readout ASIC with a shared SAR ADC for SiPM detectors," *IEEE Trans. Nucl. Sci.*, vol. 68, no. 8, pp. 2286–2293, Aug. 2021, doi: 10.1109/TNS.2021.3096162.

[138] J. Zhang, "Absolute stability of a class of neural networks with unbounded delay," *Int. J. Circuit Theory Appl.*, vol. 32, no. 1, pp. 11–21, Jan. 2004, doi: 10.1002/cta.257.

[139] H.-H. Fan, P. Cao, S.-B. Liu, and Q. An, "TOT measurement implemented in FPGA TDC," *Chin. Phys. C*, vol. 39, no. 11, Nov. 2015, Art. no. 116101, doi: 10.1088/1674-1137/39/11/116101.

[140] F. Brandonisio, A. Prodomo, M. P. Kennedy, and E. Napoli, "Implementation of a pulse-holding time-to-digital converter on an FPGA," in *Proc. Eur. Conf. Circuit Theory Design (ECCTD)*, Sep. 2013, pp. 1–4, doi: 10.1109/ECCTD.2013.6662271.

[141] T. Iizuka, T. Koga, T. Nakura, and K. Asada, "A fine-resolution pulse-shrinking time-to-digital converter with completion detection utilizing built-in offset pulse," in *Proc. IEEE Asian Solid-State Circuits Conf. (A-SSCC)*, Nov. 2016, pp. 313–316, doi: 10.1109/ASSCC.2016.7844198.

[142] C.-C. Chen, C.-S. Hwang, Y. Lin, and G.-H. Chen, "Note: All-digital pulse-shrinking time-to-digital converter with improved dynamic range," *Rev. Sci. Instrum.*, vol. 87, no. 4, pp. 1507–1510, Apr. 2016, doi: 10.1063/1.4947461.

[143] J. Zhang and D. Zhou, "An 8.5-ps two-stage Vernier delay-line loop shrinking time-to-digital converter in 130-nm flash FPGA," *IEEE Trans. Instrum. Meas.*, vol. 67, no. 2, pp. 406–414, Feb. 2018, doi: 10.1109/TIM.2017.2769239.

[144] L. Xiang, P. Yang, T. Wu, and M. Zhou, "Ultra compact pulse shrinking TDC on FPGA," *Measurement*, vol. 203, Nov. 2022, Art. no. 111874, doi: 10.1016/j.measurement.2022.111874.

[145] S. Tancock, J. Rarity, and N. Dahnoun, "The wave-union method on DSP blocks: Improving FPGA-based TDC resolutions by 3× with a 1.5× area increase," *IEEE Trans. Instrum. Meas.*, vol. 71, pp. 1–11, 2022, Art. no. 2001811, doi: 10.1109/TIM.2022.3141753.

[146] S. Tancock, J. Rarity, and N. Dahnoun, "Temperature characterisation of the DSP delay line," in *Proc. 7th Int. Conf. Event-Based Control, Commun., Signal Process. (EBCCSP)*, Jun. 2021, pp. 1–8, doi: 10.1109/EBCCSP53293.2021.9502362.

[147] Y. Hua et al., "A pulse oximeter based on time-of-flight histograms," in *Photonic Instrumentation Engineering VIII*. Bellingham, WA, USA: SPIE, 2021, doi: 10.1117/12.2577757.

[148] R. Rogalin et al., "A high dynamic-range photon-counting receiver for deep space optical communication," in *Free-Space Laser Communications XXXIII*. Bellingham, WA, USA: SPIE, 2021, doi: 10.1117/12.2576755.

[149] D. Portaluppi, K. Pasquinelli, I. Cusini, and F. Zappa, "Multi-channel FPGA time-to-digital converter with 10 ps bin and 40 ps FWHM," *IEEE Trans. Instrum. Meas.*, vol. 71, pp. 1–9, 2022, doi: 10.1109/TIM.2022.3152324.

[150] K. Hari Prasad, V. B. Chandratre, and M. Sukhwani, "An FPGA based 33-channel, 72 ps LSB time-to-digital converter," *Nucl. Instrum. Methods Phys. Res. A, Accel. Spectrom. Detect. Assoc. Equip.*, vol. 1027, Mar. 2022, Art. no. 166052, doi: 10.1016/j.nima.2021.166052.

[151] S. Moini et al., "Understanding and comparing the capabilities of on-chip voltage sensors against remote power attacks on FPGAs," in *Proc. IEEE 63rd Int. Midwest Symp. Circuits Syst. (MWSCAS)*, Aug. 2020, pp. 941–944, doi: 10.1109/MWSCAS48704.2020.9184683.

[152] Y. Zhou, Y. Wang, Z. Song, and X. Kong, "A high-precision folding time-to-digital converter implemented in Kintex-7 FPGA," *IEEE Trans. Instrum. Meas.*, vol. 72, no. 2000208, pp. 1–8, 2023, doi: 10.1109/TIM.2022.3230464.

[153] R. Szplet and A. Czuba, "Two-stage clock-free time-to-digital converter based on Vernier and tapped delay lines in FPGA device," *Electronics*, vol. 10, no. 18, p. 2190, Sep. 2021, doi: 10.3390/electronics10182190.

[154] R. Wang et al., "Picosecond timing electronics for a DIRC-like detector at the super tau-charm facility," *IEEE Trans. Nucl. Sci.*, vol. 69, no. 11, pp. 2262–2270, Nov. 2022, doi: 10.1109/TNS.2022.3211863.

[155] J. Kim, J. H. Jung, Y. Choi, J. Jung, and S. Lee, "Linearity improvement of UltraScale+ FPGA-based time-to-digital converter," *Nucl. Eng. Technol.*, vol. 55, no. 2, pp. 484–492, Feb. 2023, doi: 10.1016/j.net.2022.10.010.

[156] R. Szplet, R. Szymanowski, and D. Sondej, "Measurement uncertainty of precise interpolating time counters," *IEEE Trans. Instrum. Meas.*, vol. 68, no. 11, pp. 4348–4356, Nov. 2019, doi: 10.1109/TIM.2018.2886940.

[157] P. Chen, Y.-Y. Hsiao, Y.-S. Chung, W. X. Tsai, and J.-M. Lin, "A 2.5-ps bin size and 6.7-ps resolution FPGA time-to-digital converter based on delay wrapping and averaging," *IEEE Trans. Very Large Scale Integr. (VLSI) Syst.*, vol. 25, no. 1, pp. 114–124, Jan. 2017, doi: 10.1109/TVLSI.2016.2569626.

[158] X. Qin, L. Wang, D. Liu, Y. Zhao, X. Rong, and J. Du, "A 1.15-ps bin size and 3.5-ps single-shot precision time-to-digital converter with on-board offset correction in an FPGA," *IEEE Trans. Nucl. Sci.*, vol. 64, no. 12, pp. 2951–2957, Dec. 2017, doi: 10.1109/TNS.2017.2768082.

[159] H. Ma et al., "On-chip trust evaluation utilizing TDC-based parameter-adjustable security primitive," *IEEE Trans. Comput.-Aided Design Integr. Circuits Syst.*, vol. 40, no. 10, pp. 1985–1994, Oct. 2021, doi: 10.1109/TCAD.2020.3035346.

[160] Y. Wang et al., "Performance analysis and IP core implementation of two high performance time-to-digital converters on Xilinx 7-series FPGA," *Nucl. Instrum. Methods Phys. Res. A, Accel. Spectrom. Detect. Assoc. Equip.*, vol. 1020, Dec. 2021, Art. no. 165866, doi: 10.1016/j.nima.2021.165866.

[161] A. Banerjee et al., "Analog front-end for FPGA-based readout electronics for scintillation detectors," *Nucl. Instrum. Methods Phys. Res. A, Accel. Spectrom. Detect. Assoc. Equip.*, vol. 1028, Apr. 2022, Art. no. 166357, doi: 10.1016/j.nima.2022.166357.

[162] D. Li, R. Ma, X. Wang, J. Hu, and Z. Zhu, "Optimization of system design and calibration algorithm for SPAD-based LiDAR imager," *IEEE Trans. Instrum. Meas.*, vol. 71, pp. 1–10, 2022, Art. no. 2006010, doi: 10.1109/TIM.2022.3207801.

[163] A. Y. Barnyakov et al., "Progress and perspectives of FARICH R&D for the super charm-tau factory project," *Nucl. Instrum. Methods Phys. Res. A, Accel. Spectrom. Detect. Assoc. Equip.*, vol. 1039, Sep. 2022, Art. no. 167044, doi: 10.1016/j.nima.2022.167044.

[164] R. T. Siecha, G. Alemu, J. Prinzie, and P. Leroux, "5.7 ps resolution time-to-digital converter implementation using routing path delays," *Electronics*, vol. 12, no. 16, p. 3478, Aug. 2023, doi: 10.3390/electronics12163478.

[165] E. Zhou et al., "An FPGA-based 500 m samples per second throughput time-to-digital converter without ineffective bin," *Nucl. Instrum. Methods Phys. Res. A, Accel. Spectrom. Detect. Assoc. Equip.*, vol. 1053, Aug. 2023, Art. no. 168366, doi: 10.1016/j.nima.2023.168366.

Haojie Xia (Member, IEEE) was born in Suzhou, Anhui, China, in 1979. He received the Ph.D. degree from Hefei University of Technology, Hefei, China, in 2006.

He was a Visiting Scholar at Physikalisch-Technische Bundesanstalt (PTB), Brunswick, Lower Saxony, Germany, in 2015. He is currently a Professor at Hefei University of Technology. He mainly engages in research work in fields, such as optoelectronic precision measurement technology, micro/nano measurement and control systems, instrument accuracy theory, and precision instrument design.

Xin Yu was born in Xi'an, Shaanxi, China, in 1998. He received the B.S. degree from Xi'an University of Technology, Xi'an, in 2020. He is currently pursuing the Ph.D. degree with Hefei University of Technology, Hefei, China.

He mainly engages in research on field-programmable gate array (FPGA)-based time-to-digital converter (TDC) technology and error compensation for output orthogonal signals in optical micro/nano measurement systems.

Jin Zhang received the B.E. degree from Hefei University of Technology, Hefei, China, in 2005, and the M.S. and Ph.D. degrees from Tianjin University, Tianjin, China, in 2007 and 2010, respectively.

Since 2019, he has been a Professor with the School of Instrument Science and Optoelectronics Engineering, Hefei University of Technology. His current research interests include optical measurement technology and vibration testing.

Dr. Zhang is a member of the Youth Committee of China Instrument and Control Society. He was a recipient of the Second Prize of the Science and Technology Award of China Instrument and Control Society.

Guiping Cao received the Ph.D. degree from the University of Science and Technology of China, Hefei, China, in 2012.

He is currently the Research and Development Director of Hefei I-TEK OptoElectronics Company Ltd., Hefei, Anhui, and the Director of Anhui Provincial Key Laboratory of Precision Visual Perception. His research interests include high-speed and high-precision data acquisition systems since graduation.